

# Projection-specific integration of convergent thalamic and retrosplenial signals in the presubicular head direction cortex

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## eLife Assessment

This **valuable** study combines anatomical tracing and slice physiology to examine how anterior thalamic and retrosplenial inputs converge in the presubiculum, a key region in the navigation circuit. The authors show that near-simultaneous co-activation of retrosplenial and thalamic inputs drives supra-linear presubiculum responses, revealing a potential cellular mechanism for anchoring the brain's head direction system to external visual landmarks. Their thorough experimental approach and analyses provide **convincing** evidence for the cellular basis of how the brain's internal compass may be anchored to the external world, laying the groundwork for future experimental testing in vivo.

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## Abstract

## Summary

Head-direction (HD) signals function as the brain's internal compass. They are organized as an attractor, and anchor to the environment via visual landmarks. Here we examine how thalamic HD signals and visual information from the retrosplenial cortex combine in the presubiculum. We find that monosynaptic excitatory connections from anterior thalamic nucleus and from retrosplenial cortex converge on single layer 3 pyramidal neurons in the dorsal portion of mouse presubiculum. Independent dual wavelength photostimulation of these inputs in slices leads to action potential generation preferentially for near-coincident inputs, indicating that layer 3 neurons can transmit a visually matched HD signal to medial entorhinal cortex. Layer 4 neurons, which innervate the lateral mammillary nucleus, form a second step in the association of HD and landmark signals. They receive little direct input from thalamic and retrosplenial axons. We show that layer 4 cells are excited di-synaptically, transforming regular spiking activity into bursts of action potentials, and that their firing is

enhanced by cholinergic agonists. Thus, a coherent sense of orientation involves projection specific translaminar processing in the presubiculum, where neuromodulation facilitates landmark updating of HD signals in the lateral mammillary nucleus.

## Introduction

The head direction (HD) system functions as the brain's compass system. It is distributed across several interconnected brain structures, and hierarchically organized from the brainstem to the lateral mammillary nucleus (LMN; [Stackman and Taube, 1998](#)), the anterior thalamic nuclei (ATN; [Taube, 1995](#)), and the dorsal Presubiculum (PrS; [Ranck, 1984](#); [Taube et al., 1990a](#)). Modeling work suggests the HD system functions as a ring attractor, where the population of HD neurons is arranged in a one dimensional manifold ([Blair and Sharp, 1995](#); [McNaughton et al., 2006](#); [Skaggs et al., 1995](#)). The activity of thalamic and PrS HD neurons is correlated across different brain states, even during sleep, independently of sensory cues ([Peyrache et al., 2015](#)). The bump-like activity dynamics of an attractor network can maintain HD signals based on excitatory-excitatory or excitatory-inhibitory interactions ([Knierim and Zhang, 2012](#); [Simonnet et al., 2017](#)). However, HD signals may drift in the dark ([Taube et al., 1990b](#); [Zugaro et al., 2003](#)). For precise navigation, internally generated information on head direction must be combined with awareness of location in an environment. Mechanisms underlying the combination of egocentric head direction signals with anchoring to allocentric landmarks remain to be clarified.

Head direction and visual landmark signals may be integrated in the presubiculum ([Jeffery et al., 2016](#); [Yoder et al., 2019](#)). Head direction cells are found in the superficial and deep layers of dorsal presubiculum, also termed postsubiculum ([Boccarda et al., 2010](#)), and most layer 3 neurons in the dorsal presubiculum are head direction cells ([Tukker et al., 2015](#); [Preston-Ferrer et al., 2016](#)). These cells receive monosynaptic head direction inputs from the anterior thalamus ([Nassar et al., 2018](#); [Peyrache et al., 2015](#); [Balsamo et al., 2022](#)). Lesions of the PrS impair the visual landmark control of a cell's preferred direction in ATN and in LMN ([Goodridge and Taube, 1997](#); [Yoder et al., 2015](#)). Presubicular lesions also induce place field instability in the hippocampus ([Calton et al., 2003](#)), suggesting this region may be crucial to the anchoring of directionally modulated neurons to environmental landmarks.

Landmark based navigation depends on reliable visual cues. The PrS appears to receive direct projections from the primary visual cortex and indirect visual input via the retrosplenial cortex ([Van Groen & Wyss 2003](#); [Vogt & Miller 1983](#)). The retrosplenial cortex encodes angular head velocity ([Alexander & Nitz 2015](#); [Keshavarzi et al. 2022](#)) and visual landmark information ([Auger et al., 2012](#); [Clark et al., 2010](#); [Sit and Goard, 2023](#)), and supports an array of cognitive functions including memory, spatial and non-spatial context and navigation ([Vann et al., 2009](#); [Vedder et al., 2017](#)). In particular, some neurons in the dysgranular retrosplenial cortex encode landmark-dominated head-direction signals ([Jacob et al., 2017](#)). Presubicular layer 3 cells receiving projections from both the anterior thalamus and the retrosplenial cortex ([Kononenko and Witter 2012](#)) could update the compass and bind landmarks to head direction signals. Presubicular neurons could then broadcast integrated HD-landmark signals directly to the MEC, via deep layers to the ADN and via layer 4 neurons to the LMN ([Huang et al., 2017](#); [Yoder et al., 2015](#); [Yoder and Taube, 2011](#)). The different target-specific presubicular projection neurons are well positioned to integrate anterior thalamic and retrosplenial inputs, but technical constraints have limited understanding of how known anatomical inputs are transformed into functional integrated output signals.

This work was therefore designed to examine the convergence of visual landmark information, that is possibly integrated in the RSC, and vestibular based thalamic head direction signals, in the PrS. Retrograde tracing was used to confirm inputs to the PrS from the ATN and the RSC. The

spatial distribution of ATN and RSC targets in the dorsal and ventral PrS was investigated by stereotaxic injection of viral vectors inducing anterograde expression of light-gated ion channels fused to fluorescent reporters. We found that superficial layers of the dorsal PrS are major targets of ATN and RSC projections. We analyzed functional convergence of these inputs in the PrS using dual-wavelength optogenetic stimulations in ex vivo brain slices, while recording from layer 3 and 4 pyramidal neurons. Both ATN and RSC projections made mono-synaptic excitatory connections with single layer 3 principal cells and mostly di-synaptic ones with layer 4 cells. We show that EPSPs induced with close to coincident timing by ATN and RSC fibers summed non-linearly in layer 3 neurons. Layer 4 cell firing is facilitated by cholinergic activation. These data provide insights into the integration of landmark and head direction inputs and their distribution to downstream targets by PrS pyramidal cells.

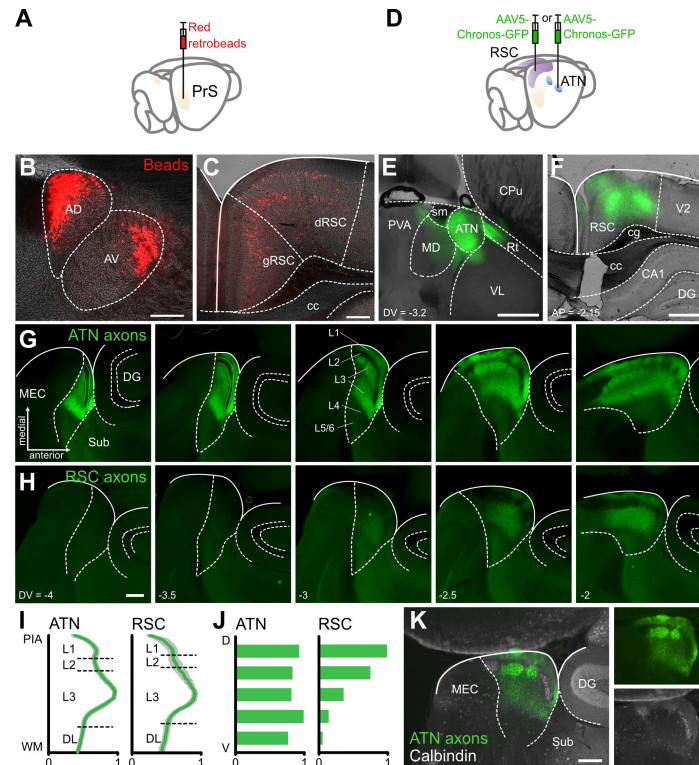
## Results

### ATN and RSC send strong axonal projections to the dorsal Presubiculum

The origins of the main afferents projecting to the presubiculum were explored by injecting retrogradely transported fluorescent beads into the PrS (**Figure 1A**). After 4 days, coronal slices were prepared to examine the injection site and transport of the beads. Some beads remained close to the PrS injection site, due in part to local projections. Strong bead signals were detected within two of the subnuclei that form the anterior thalamus, the anterodorsal (AD) and anteroventral (AV) part of ATN. Neurons in AD were labeled most strongly in the medial portion of the AD, while labeled neurons in AV were found in its lateral portion (**Figure 1B**). Many neurons in the RSC were labeled. Cells with somata in layers 2 and 5 of the dysgranular dRSC were labeled, while mostly layer 5 cells of granular gRSC contained beads (**Figure 1C**). Regions adjacent to the presubiculum including the subiculum (Sub), parasubiculum (PaS) and the medial and lateral entorhinal cortices (MEC, LEC) were labeled as was the contralateral PrS. Beads were also detected in the laterodorsal thalamic nucleus (LD). **Figure 1—figure supplement 1** shows an example of a series of labeled coronal sections from one out of two mice, indicating some labeling also in the visual cortices, perirhinal cortex, the nucleus reuniens of the thalamus, the dorsolateral geniculate nucleus, and the claustrum. Some beads were observed in the CA1 region, which has not previously been reported. Potentially though, this labeling could derive from a bead leak into the nearby PaS which is innervated by CA1 (Van Groen et Wyss 1990a). In summary, the ATN and RSC are the major sources of afferents projecting to the presubiculum with lesser inputs from other sites.

Projections from the ATN or RSC to the presubiculum were explored by injecting at these sites an anterogradely transported viral construct which expressed the modified Channelrhodopsin Chronos fused to GFP (**Figure 1D**). After 4 weeks, horizontal slices were prepared to verify the ATN injection site. Chronos-GFP labeling was mostly confined to the ATN, occasionally extending to medial thalamic and reticular nuclei nearby, possibly as bundles of projecting fibers (**Figure 1E**). After injection at the RSC site, coronal sections (**Figure 1F**) showed strong expression in layer 5, especially deeper zones, and layer 6 of both dysgranular and granular RSC, while labeled dendrites were evident in layer 1. Chronos-GFP expression extended throughout the RSC, from -1.7 to -3.7 posterior to Bregma on the antero-posterior axis.

Projections of the ATN to the presubiculum were examined in horizontal sections (**Figure 1G**). ATN axons expressing the Chronos-GFP construct targeted layers 1 and 3 of the presubiculum, mostly avoiding layer 2 (Simonnet et al., 2017; Nassar et al., 2018). Labeling was precisely limited to the presubiculum with a sharp reduction of green fluorescence at the posterior border with the parasubiculum, the anterior border with the subiculum and the lateral border with deep layers. Axon terminals in layer 3 were not homogeneously distributed. Patches of higher density



**Figure 1.**

### ATN and RSC send strong axonal projections to the superficial layers of dorsal Presubiculum

A, Retrograde labeling of cortical and subcortical regions projecting to the presubiculum (PrS) with retrobeads.

B, Ipsilateral anterodorsal and anteroventral thalamic nuclei are labeled with beads.

C, Ipsilateral granular (gRSC) and dysgranular (dRSC) retrosplenial cortex labeling. Scale bars B, C, 200  $\mu$ m.

D, Anterograde labeling of thalamic and retrosplenial projections to PrS with AAV-Chronos-GFP.

E, F, Injection sites in ATN (E) and RSC (F). AD: anterodorsal thalamic nucleus, AV: anteroventral thalamic nucleus, CA1: field of the hippocampus, cc: corpus callosum, cg: cingulum, CPu: caudate putamen, DG: dentate gyrus, MD: thalamic medial dorsal nucleus, MEC: medial entorhinal cortex, PaS: parasubiculum, PVA: paraventricular thalamic nucleus, anterior, Rt: thalamic reticular nucleus, sm: stria medullaris, Sub: subiculum, VL: thalamic ventrolateral nucleus.

G, H, Five sequential 100- $\mu$ m horizontal slices of the parahippocampal region show ATN (G) and RSC (H) axons expressing Chronos-GFP (green). Dashed lines show limits of the parahippocampal region to the left and of the dentate gyrus to the right. Numbers show dorsoventral level with respect to bregma. Scale bars 100  $\mu$ m.

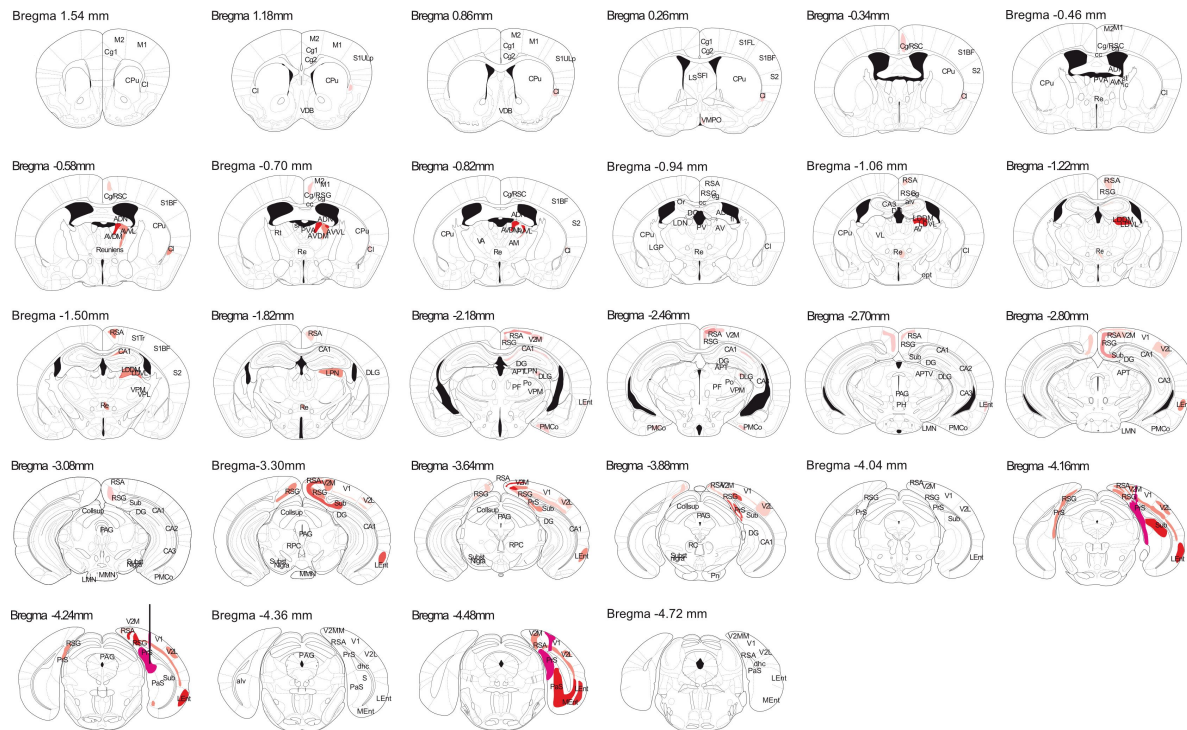
I, Normalized profiles of fluorescent intensity for ATN and RSC projections to the presubiculum, from white matter (WM) to pia (PIA). Mean (green)  $\pm$  SEM (grey),  $n = 8$  slices each.

J, Ventral (V) to dorsal (D) normalized distribution of ATN and RSC projections to the presubiculum,  $n = 3$  mice.

K, ATN axon labeling (green) is segregated from calbindin labeling (white) in the PrS. Scale bar 100  $\mu$ m.

See also **Figure 1—figure supplement 1**.



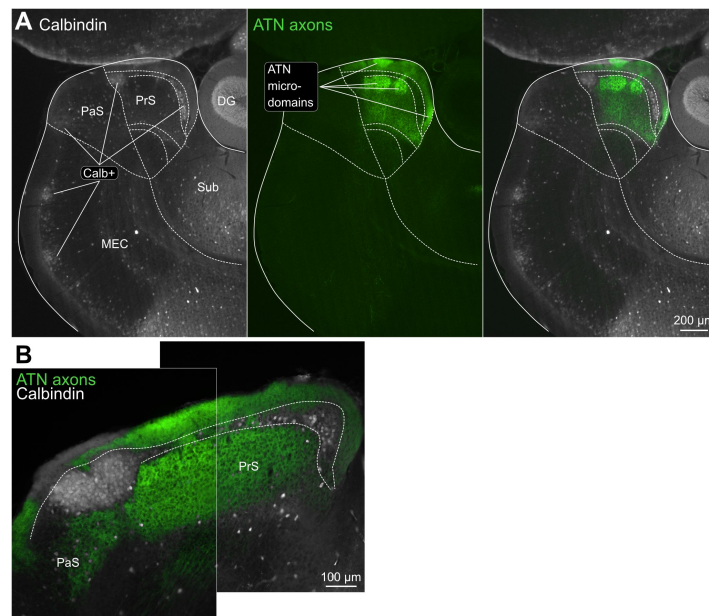


**Figure 1—figure supplement 1.**

### Brain regions providing input to the Presubiculum.

Semi-quantitative estimate of retrobeads labelling in serial coronal slices following injection in the presubiculum. Color intensity represents beads density (mouse #69).

- Thalamic nuclei: anterodorsal (AD) +++, anteroventral (AV) +++, laterodorsal (LD) +++, lateroposterior ++, dorsal lateral geniculate ++, reunions ++, anteromedial -, ventrolateral -
  - Retrosplenial cortex: dysgranular (dRSC) +++, granular +++, dysgranular contralateral ++
  - Visual cortex: primary (V1) ++, secondary (V2L) +++
  - Entorhinal cortex: lateral (LEnt) +++, medial (MEnt) +++
  - Hippocampus/parahippocampus: parasubiculum (PaS) +++, subiculum (Sub) +++, contralateral PrS ++, CA1 ++, CA3 -, dentate gyrus -
  - Other: claustrum ++, perirhinal cortex ++, temporal association cortex ++
- ADN: anterodorsal thalamic nucleus; AM: anteromedial thalamic nucleus; APT(V): anterior pretectal nucleus (ventral part); AVDM: anteroventral thalamic nucleus, dorsomedial part; AVVL: anteroventral thalamic nucleus, ventrolateral part; CA: cornu ammonis; cc: corpus callosum; Cl: Claustrum; Cg1,2: cingular cortex area 1,2; coll. sup: colliculus superior; CPU: caudate putamen (striatum); dhc: dorsal hippocampal commissure; DG: dentate gyrus; DLG: dorsal lateral geniculate nucleus; LEnt: lateral entorhinal cortex; LDDM: laterodorsal thalamic nucleus, dorsomedial part; LDVL: laterodorsal thalamic nucleus, ventrolateral part; LMN: lateral mammillary nucleus; LS: lateral septal nucleus; M1,2: primary, secondary motor cortex; MEnt: medial entorhinal cortex; MMN: medial mammillary nucleus; PAG: periaqueductal gray; PaS: Parasubiculum; PF: parafascicular thalamic nucleus; PH: posterior hypothalamic area; PMCo: posteromedial cortical amygdaloid nucleus; Po: posterior thalamic nuclear group; PrH: perirhinal cortex; PrS: presubiculum; PVA: paraventricular thalamic nucleus, anterior part; S1BF, S1FL, S1ULp, S1Tr: primary somatosensory cortex, barrel field, forelimb region, upper lip region, trunk region; S2: secondary somatosensory cortex; sm: stria medullaris of the thalamus; Sub: subiculum; Subst. Nigra: substantia nigra; Re/Reunions: reunions thalamic nucleus; RSA = dRSC, dysgranular retrosplenial cortex; RSC: retrosplenial cortex; RSG = gRSC: granular retrosplenial cortex; Rt: reticular thalamic nucleus; V1: primary visual cortex; V2M: secondary visual cortex, medial part; V2L: secondary visual cortex, lateral part; VA: ventral anterior thalamic nucleus; VDB: nucleus of the vertical limb of the diagonal band; VL: ventrolateral thalamic nucleus; VPL: ventral posterolateral thalamic nucleus; VPM: ventral posteromedial thalamic nucleus



**Figure 1—figure supplement 2.**

**Calbindin immunostaining and GFP-expressing ATN axons in horizontal sections.**

A, Subicular (Sub), Parasubicular (PaS), medial entorhinal cortical (MEC), and presubicular (PrS) borders are outlined, as well as the dentate gyrus (DG). Patches of high density of ATN axonal ramifications in L3 are indicated as "ATN microdomains". B, Territories devoid of ATN axons in PrS L2 stain positive for calbindin.

were evident in deep layer 3 (**Figure 1G** [↗](#), left panel) or in the upper layer 3 (**Figure 1G** [↗](#), forth panel, and **Figure 1K** [↗](#)). Along the dorso-ventral axis, ATN axons projected from -2 to -4 mm ventral to Bregma. Fluorescence was not detected in adjacent regions including the Sub, PaS, and DG. ATN axons avoided layer 2 of the PrS, including regions with patches of calbindin positive neurons that project to contralateral PrS (Preston-Ferrer et al. 2016 [↗](#); **Figure 1K** [↗](#)).

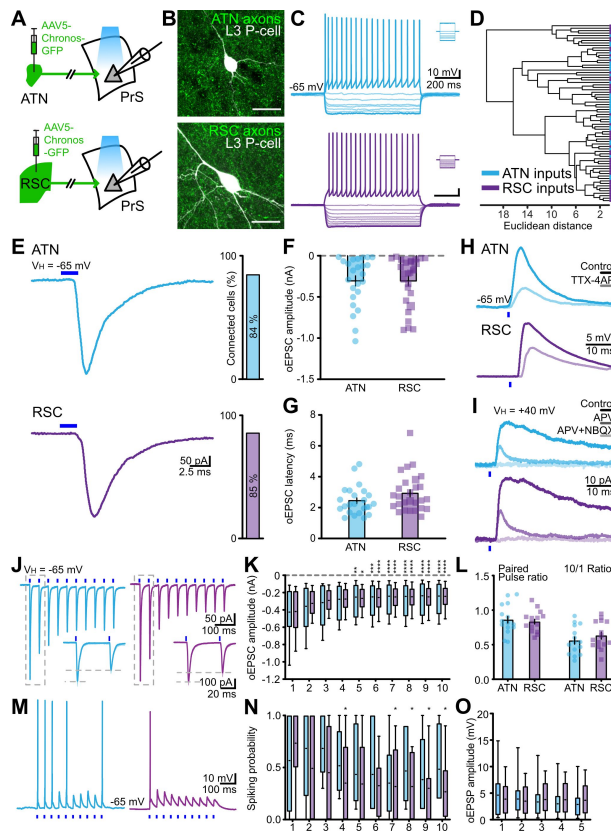
RSC projections to the presubiculum (**Figure 1H** [↗](#)), traced with the fluorescent Chronos-GFP construct, also projected to layers 1 and 3. Unlike ATN axons, they were restricted to dorsal regions of the presubiculum (from -2 to -3 mm ventrally to Bregma on the dorso-ventral axis; **Figure 1H, I, J** [↗](#)). RSC axons also presented a patchy innervation pattern in L1-3, and tended to avoid microdomains of high-density ATN projections (**Figure 1G, K** [↗](#), **Figure 3B** [↗](#)). Overall, these data show ATN and RSC afferents innervate overlapping dorsal regions of the PrS, in layer 1 and parts of layer 3. Different spatial patterns of fiber terminations suggest distinct local connectivities.

## Layer 3 neurons are directly excited by both ATN and RSC afferents

We next examined physiological effects of ATN and RSC afferents on PrS pyramidal cells (Rees et al., 2017 [↗](#)). Responses to photo-stimulation of the Chronos-GFP construct injected either in the thalamus or in the retrosplenial cortex (**Figure 2A** [↗](#)) were recorded in layer 3 neurons (**Figure 2B** [↗](#)). These cells had robust regular firing patterns in response to depolarizing current injections (**Figure 2C** [↗](#)). Intrinsic properties of layer 3 cells were rather uniform (**Figure 2—figure supplement 1** [↗](#)) and an unsupervised cluster analysis suggested they formed a single population (**Figure 2D** [↗](#)).

Blue light activation of either ATN or RSC axons expressing Chronos-GFP reliably evoked synaptic responses in whole-cell patch-clamp records from layer 3 cells. High intensity stimulation (1 - 3 mW) evoked EPSCs or EPSPs with a probability of 84% (70/83 neurons from 14 mice) for ATN fibers and 85% (104/123 neurons from 20 mice) for RSC fibers (**Figure 2E** [↗](#)). Excitatory postsynaptic currents or potentials evoked by optical stimulation of thalamic and retrosplenial inputs will be referred to as oEPSCs and oEPSPs. Mean oEPSC amplitudes (**Figure 2F** [↗](#)) were similar for ATN ( $-305.0 \pm 61.0$  pA,  $n = 24$ ) and RSC fiber stimuli ( $-305.8 \pm 59.1$  pA,  $n = 27$ ). Response latency (**Figure 2G** [↗](#)) to ATN axon stimulation was  $2.45 \pm 0.19$  ms ( $n = 24$ ) and that for RSC axon stimulation was  $2.93 \pm 0.23$  ms ( $n = 27$ ). The variability in latency of oEPSCs (**Figure 2—figure supplement 2A, B** [↗](#)) induced by RSC afferent stimuli was  $0.35 \pm 0.09$  ms ( $n = 27$ ) and that for oEPSCs induced by ATN stimuli was  $0.19 \pm 0.06$  ms ( $n = 24$ ). These short oEPSC latencies suggest monosynaptic transmission. We confirmed this point by showing that oEPSCs were maintained in presence of TTX (1  $\mu$ M) and 4AP (100  $\mu$ M) (**Figure 2H** [↗](#),  $n = 2$  and 12 for ATN and RSC respectively). EPSCs induced by stimulation of ATN and RSC fibers were reduced in amplitude and decayed more quickly in the presence of the NMDA receptor antagonist APV (50  $\mu$ M) and were completely abolished by further application of the AMPA receptor antagonist NBQX (1 mM) (**Figure 2I** [↗](#)). The shape of oEPSCs induced by light stimulation of ATN and RSC fibers was comparable. Rise times were  $1.38 \pm 0.15$  ms for ATN EPSCs ( $n = 15$ ) and  $1.38 \pm 0.08$  ms for RSC-mediated EPSCs ( $n = 15$ ) and mean half-widths were  $4.00 \pm 1.02$  ms for ATN-induced EPSCs ( $n = 15$ ) and  $4.34 \pm 1.18$  ms for RSC EPSCs ( $n = 15$ ). The mean time constant (tau decay) for ATN-mediated EPSCs was  $3.01 \pm 0.21$  ( $n = 15$ ) and for RSC-induced EPSCs  $3.17 \pm 0.27$  ( $n = 15$ ) (**Figure 2—figure supplement 2C-E** [↗](#)).

Optical stimulation of ATN and RSC fibers at lower light intensities elicited sub-threshold synaptic events in layer 3 cells. oEPSP amplitude for ATN mediated events was  $4.09 \pm 1.18$  mV ( $n = 11$ ) and for RSC events it was  $4.80 \pm 0.84$  mV ( $n = 11$ ). Maximal rising slopes were  $2.93 \pm 0.82$  mV/ms for ATN-mediated events ( $n = 11$ ) and  $3.02 \pm 0.53$  mV/ms for RSC EPSPs ( $n = 11$ ), and decay time constants were  $81.4 \pm 11.6$  ms for ATN EPSPs ( $n = 11$ ) and  $86.9 \pm 11.2$  ms for RSC EPSPs ( $n = 11$ ). Rise times were shorter for subthreshold EPSPs induced by ATN afferents at  $3.00 \pm 0.24$  ms ( $n = 11$ ) while those for RSC initiated EPSPs were  $3.59 \pm 0.34$  ( $n = 11$ ,  $p = 0.042$ , Wilcoxon test; **Figure 2—figure supplement 2F-J** [↗](#)). These data show that ATN and RSC axons make monosynaptic, glutamatergic excitatory contacts on layer 3 PrS cells.



**Figure 2.**

### Responses of layer 3 presubicular cells to photoactivation of ATN and RSC fibers.

A, Expression of AAV5-Chronos in ATN or RSC.

B, Biocytin labeled layer 3 cells (white) and GFP positive axons (green) from the ATN or RSC. Scale bar 10  $\mu$ m.

C, Firing pattern of two layer 3 cells receiving inputs from the ATN (*cerulean*) and the RSC (*purple*). Insets show current commands.

D, Cluster analysis of physiological parameters for cells tested by stimulating ATN (n = 29 cells /14 mice) or RSC (n = 43 cells/20 mice) fibers.

E, Representative EPSCs evoked in layer 3 cells by light stimulation (blue bar) of ATN or RSC inputs. Right, proportion of cells receiving ATN or RSC inputs.

F, Average amplitudes of ATN and RSC induced synaptic currents. Mann-Whitney U test revealed no significant difference ( $p = 0.95$ ).

G, Latency of EPSCs evoked by light-stimulation of ATN (n = 24 cells/12 mice) or RSC fibers (n = 27 cells/14 mice). Mann-Whitney U test revealed no significant difference ( $p = 0.13$ ).

H, EPSPs induced in layer 3 cells (single traces) by stimulating ATN or RSC inputs in absence and presence of 1  $\mu$ M TTX and 100  $\mu$ M 4-AP.

I, EPSCs induced by stimulating ATN or RSC fibers in the absence and presence of 100  $\mu$ M APV and APV + 10  $\mu$ M NBQX. Holding potential +40 mV.

J, Voltage-clamp responses of layer 3 cells to 20 Hz train stimulations of ATN (*left*) and RSC (*right*) inputs. Insets show EPSCs in response to the first two stimuli.

K, oEPSC amplitudes for 10 ATN or RSC fiber stimuli at 20 Hz (ATN, n = 15 cells/8 mice, RSC, n = 15 cells/11 mice). Short middle line, mean; Min to max and quartiles.

L, Paired-pulse ratio (PPR) and 10/1 ratio (ratio between 10<sup>th</sup> and 1<sup>st</sup> event amplitudes) for ATN or RSC inputs. Wilcoxon matched-pairs signed rank test: ATN PPR vs 10/1 \*\*\*\*  $p < 0.0001$ , RSC PPR vs 10/1 \*\*\*  $p = 0.0001$ .

M, Current clamp traces showing action potentials and EPSPs evoked by 10 stimuli at 20 Hz. N, Spiking probability during trains of 10 stimuli. ATN, n = 6 cells/4 mice, RSC, n = 12 cells/9 mice. Full line, median; short line, mean; Min to max and quartiles. In K and N, \*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$ , \*\*\*\*  $p < 0.0001$  from Friedman's test followed by Dunn's *post-hoc* test.

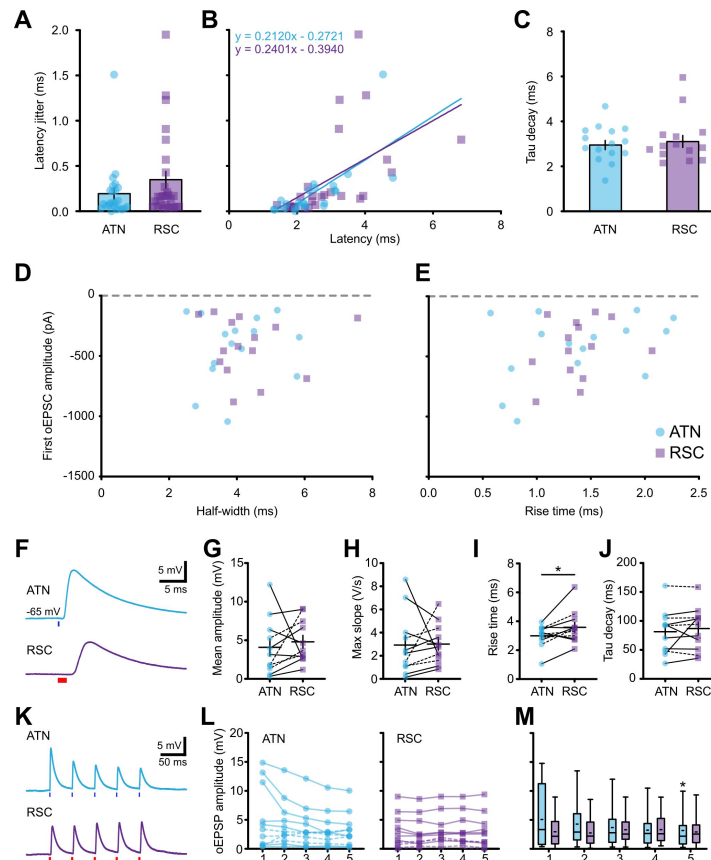
O, oEPSP amplitudes for trains of 5 stimuli at 20 Hz. See also Figure 2—figure supplements 1 and 2.

|                                      | ATN    |      |    | RSC    |      |    | <i>p-value</i> |
|--------------------------------------|--------|------|----|--------|------|----|----------------|
|                                      | Mean   | SEM  | n  | Mean   | SEM  | n  | Mann-Whitney   |
| Resting membrane potential (mV)      | -71.37 | 1.85 | 27 | -73.50 | 1.52 | 38 | <i>ns</i>      |
| Neuronal input resistance (MΩ)       | 414.5  | 31.5 | 27 | 349.8  | 23.4 | 38 | <i>ns</i>      |
| Tau 1 (ms)                           | 26.46  | 1.80 | 27 | 22.25  | 1.19 | 38 | <i>0.0481</i>  |
| Sag ratio at -100 mV                 | 1.07   | 0.01 | 27 | 1.10   | 0.01 | 38 | <i>ns</i>      |
| Rheobase current (pA)                | 51.88  | 5.25 | 27 | 63.00  | 3.67 | 38 | <i>0.0188</i>  |
| Firing rate at 200 pA (Hz)           | 39.21  | 3.87 | 27 | 45.10  | 3.02 | 38 | <i>ns</i>      |
| Maximum firing frequency (Hz)        | 48.19  | 4.56 | 27 | 64.84  | 5.02 | 38 | <i>0.0035</i>  |
| Input-output slope (Hz/nA)           | 357.5  | 30.0 | 27 | 376.7  | 24.6 | 38 | <i>ns</i>      |
| AP threshold (mV)                    | -34.13 | 0.86 | 27 | -33.48 | 0.91 | 38 | <i>ns</i>      |
| AP width (ms)                        | 0.77   | 0.06 | 27 | 0.68   | 0.04 | 38 | <i>ns</i>      |
| AP AHP (mV)                          | -20.25 | 0.64 | 27 | -18.77 | 0.47 | 38 | <i>ns</i>      |
| AP rise amplitude (mV)               | 77.84  | 1.67 | 27 | 79.77  | 1.26 | 38 | <i>ns</i>      |
| AP maximum depolarization rate (V/s) | 323.7  | 11.0 | 27 | 350.7  | 9.2  | 38 | <i>0.0326</i>  |
| AP maximum repolarization rate (V/s) | -107.9 | 6.1  | 27 | -121.7 | 5.9  | 38 | <i>ns</i>      |
| Onset latency at rheobase (ms)       | 181.2  | 24.3 | 27 | 139.4  | 17.6 | 38 | <i>ns</i>      |

**Figure 2—figure supplement 1.**

**Table showing the intrinsic electrophysiological physiological properties of layer 3 cells tested for responses to stimulating ATN or RSC fibers.**





**Figure 2—figure supplement 2.**

### Comparison of evoked synaptic events in presubicular layer 3 neurons following photostimulations of ATN or RSC axons.

A, Variance of latency (jitter) in oEPSP onset following Chronos-expressing ATN or RSC fiber stimuli. RSC input stimulation gives similar variation in latency as ATN inputs. Mann-Whitney U test revealed no significant difference ( $p = 0.26$ ). B, Jitter as a function of latency for ATN and RSC stimuli shows a similar coefficient of variation. C, Decay time constants of EPSCs evoked in layer 3 neurons by stimulating ATN or RSC inputs. Mann-Whitney U test revealed no significant difference ( $p = 0.87$ ). D, E, First oEPSC amplitude as a function of oEPSC half-width (D) and oEPSC rise time (E) for both inputs. We found no clear segregation of oEPSC shapes according to the origin of afferent fibers. F, oEPSPs in a presubicular layer 3 cell, induced by blue or red light stimulation of Chronos-expressing ATN or Chrimson-expressing RSC afferents, respectively. G-J, Comparison of oEPSPs evoked by stimulating ATN or RSC axons, recorded from a same postsynaptic layer 3 presubicular neuron. G, Mean amplitude. H, Maximum rising slope of oEPSPs. I, oEPSPs rise time.  $p < 0.05$ , Wilcoxon test. J, oEPSPs decay time constants. oEPSPs following ATN and RSC fiber stimulation recorded from the same cell are connected by lines. Dashed lines indicate reversed experimental design (expressing Chronos in RSC and Chrimson in ATN). K, Responses to 10 light stimuli at 20 Hz for ATN and RSC inputs to the same cell. L, Amplitudes of oEPSPs evoked by trains of ATN (left) or RSC (right) fiber stimuli for 11 cells. Dashed lines indicate the reversed experimental design (expressing Chronos in RSC and Chrimson in ATN). M, Summary data from L. On average, the amplitudes of oEPSPs evoked by ATN fiber stimulation were smaller for the fifth than for the first stimulus. oEPSPs evoked by RSC fiber stimulation were similar across five stimuli. Full line, median; Short line, mean; Min to max and quartiles. \*  $p < 0.05$ , Friedman's test and post-hoc Dunn's test comparing each stimulus with the first.

The dynamics of responses to repetitive stimulation of layer 3 PrS cells were tested using 20 Hz trains of light stimuli to activate either ATN or RSC afferents. oEPSCs for both inputs followed depressing kinetics. Amplitude decreased for both inputs (**Figure 2J, K**), significantly after the fourth pulse (Friedman's test and Dunn's multiple comparison test). The 10/1 ratio was significantly lower than the paired-pulse ratio (PPR) in both cases (ATN 10/1  $0.56 \pm 0.05$  vs PPR  $0.86 \pm 0.05$ ,  $n = 15$ ,  $p < 0.0001$ , Wilcoxon test, RSC 10/1  $0.64 \pm 0.05$  vs PPR  $0.83 \pm 0.03$ ,  $n = 15$ , Wilcoxon test) (**Figure 2L**). ATN fiber stimulation induced PrS cell firing in current-clamp with a high probability for initial stimuli and lower stable probabilities for later stimuli in the train. Similarly, 20 Hz stimulation of RSC afferents evoked spikes with high probability for initial stimuli and decreasing probabilities for later stimuli (on fourth and 7<sup>th</sup>-10<sup>th</sup> pulses, Friedman's and Dunn's multiple comparison tests; **Figure 2M, N**). Responses to repeated subthreshold stimuli had similar dynamics (**Figure 2O**). Stimulation with lower light intensities ( $< 1$  mW) to initiate subthreshold oEPSPs elicited different dynamic responses. ATN afferent trains elicited responses with depressing dynamics: the amplitude of the fifth oEPSP was significantly less than the first one ( $3.18 \pm 0.88$  vs  $5.11 \pm 1.64$  mV,  $p = 0.0185$ , Friedman's and Dunn's tests). In contrast, subthreshold 20 Hz stimulation of RSC afferents evoked little or no oEPSP depression (**Figure 2—figure supplement 2K-M**). These data reveal distinct patterns of integration of repetitive activity in glutamatergic ATN and RSC afferents terminating on PrS layer 3 pyramidal cells.

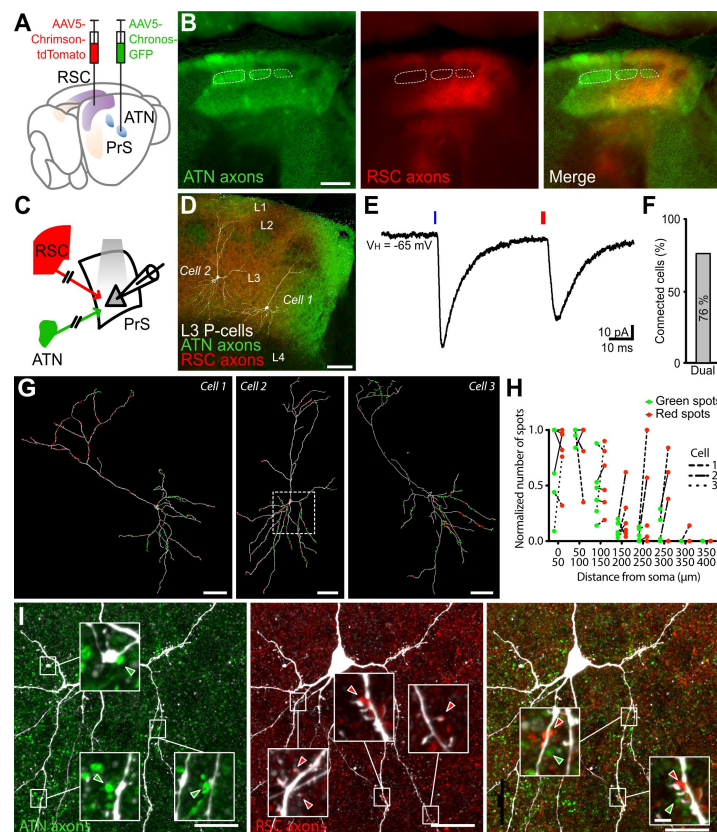
## Convergence of ATN and RSC inputs on single layer 3 neurons

Are single layer 3 pyramidal cells innervated by both ATN and RSC afferents? It seemed likely since the probability of connection to a given cell was 84% for ATN fibers and 85% for RSC afferents, and cluster analysis (**Figure 2D**) provided no evidence for subpopulations of layer 3 cells in terms of intrinsic electrophysiological properties (see also [Balsamo et al., 2022](#)).

We tested the hypothesis using photostimulation at two different light frequencies to activate ATN and RSC afferents independently. However this approach may be compromised since there is an overlap in the excitation spectra of blue-light activated Chronos (400-600 nm) and red-light activated Chrimson (450-700 nm; [Klapoetke et al., 2014](#)). Excitation of Chronos with high intensities of blue light might also excite Chrimson. Stimulating with red light at 630 nm should elicit neurotransmitter release in Chrimson-containing but not Chronos-expressing fibers, which we found was indeed the case. Blue light stimuli at 470 nm and intensities higher than 0.01 mW elicited oEPSPs in layer 3 cells of animals with Chronos-expressing fibers. In contrast, blue light intensities higher than 0.25 mW were needed to induce oEPSPs in different animals with Chrimson-expressing axons (**Figure 3—figure supplement 1A-D**). Moreover, the amplitude of these events was smaller in Chrimson-injected than in Chronos-injected mice. Thus, in our experimental conditions, blue light intensities up to 0.25 mW could be used to excite Chronos-positive axons with confidence that Chrimson-expressing fibers would not be stimulated.

We also tested the inverse, red light component of the dual stimulus strategy. Intensities were tested in mice injected either with AAV5-Chronos to label ATN fibers or AAV5-Chrimson to label RSC afferents. The strategy was validated by showing that red light induced oEPSPs in Chrimson-containing, but not Chronos-expressing fibers, while blue light evoked synaptic events in Chronos-, but not Chrimson-expressing fibers (**supplement 1E-K**).

Injection in the same animal of the Chronos construction to the thalamus and that for Chrimson to the retrosplenial cortex let us visualize ATN and RSC afferents to the presubiculum (**Figure 3A, B**). Axon terminal fields of ATN and RSC projections were evident in superficial layers of dorsal PrS (cf. **Figure 1**). Layer specificities, and labeling inhomogeneities were apparent as were high density ATN patches below layer 2 with no RSC axon innervation (**Figure 3B**). Layer 3 pyramidal cells could be innervated by ATN and RSC contacts both with apical dendrites in layer 1 and basilar dendrites of layer 3 (**Figure 3D**). We found optical stimulation of ATN and RSC axons, evoked synaptic responses in 14 out of 17 layer 3 pyramidal cells tested (current-clamp and voltage-clamp; **Figure 3E, F**).



**Figure 3.**

### ATN and RSC axons converge in dorsal presubiculum and contact single layer 3 pyramidal neurons.

A, Expression of the blue light sensitive opsin Chronos in ATN and the red light sensitive opsin Chrimson in RSC.

B, Axons from ATN (GFP labeled, green) and RSC (tdTomato labeled, red) overlap in layer 1 and 3 of dorsal presubiculum. Layer 2 possesses patches of axon-dense and axon-poor zones. RSC fibers avoid axon-dense microstructures formed by ATN fibers in upper layer 3. Scale bar, 200  $\mu\text{m}$ .

C, Independent dual wavelength optogenetic stimulation of light sensitive afferent fibers in presubicular slices.

D, Two biocytin labeled PrS layer 3 pyramidal cells surrounded by ATN (green) and RSC (red) axons.

E, Patch clamp recording from a layer 3 neuron shows optical EPSCs following photostimulation of ATN axons (blue light) and RSC axons (red light).

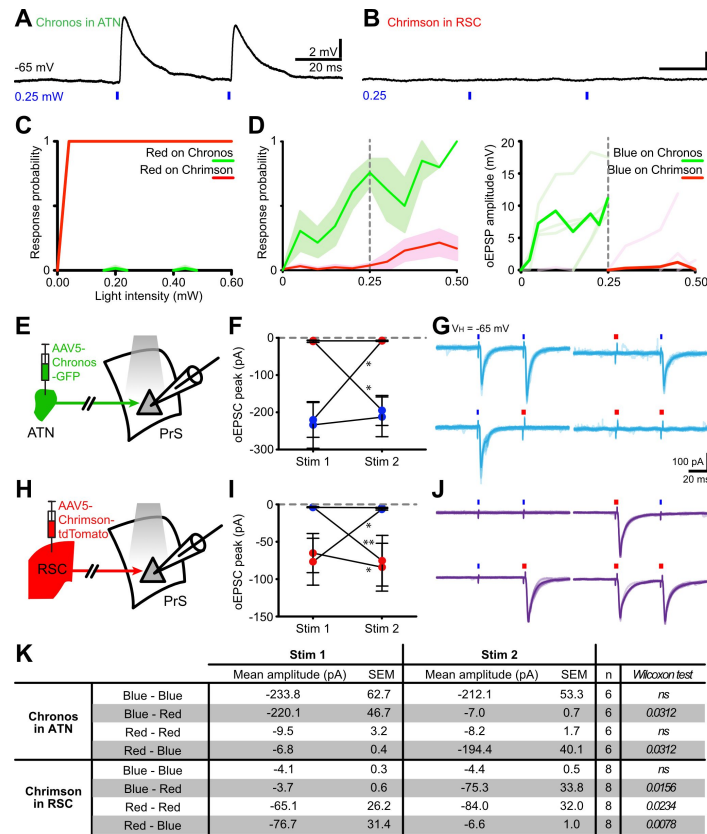
F, 76% of layer 3 pyramidal neurons tested ( $n = 17$  cells/4 mice) received both ATN and RSC input.

G, Distributions of putative synaptic contacts from ATN (green) and RSC (red) on the dendrites of three layer 3 neurons. Scale bar 50  $\mu\text{m}$ . The boxed area on Cell 2 is shown in panel I.

H, Normalized number of green and red spots for 6 neurons as a function of the distance from soma. Paired values are indicated by dotted lines for the 3 cells in G.

I, Examples of ATN-labeled (left), RSC-labeled (middle) and both (right) synapses closely apposed to dendrites of a biocytin-filled layer 3 pyramidal cell. Scale bar 20  $\mu\text{m}$ . Insets show representative high-magnification images. Scale bar 2  $\mu\text{m}$ .

See also **Figure 3—figure supplement 1**.



**Figure 3—figure supplement 1.**

### Calibration of blue and red light stimulation of Chronos and Chrimson.

A, Blue light stimulation of ATN Chronos-expressing fibers evoked oEPSPs in presubicular layer 3 neurons at an intensity of 0.25mW, and 0.5 ms duration of illumination. B, Blue light stimulation at the same intensity and duration did not evoke oEPSPs from RSC Chrimson-expressing fibers. C, D, Response probabilities of presubicular layer 3 cells to stimulation of fibers expressing Chronos or Chrimson with red (C) or blue (D, left) light. E, Experimental design. Injection of AAV5-Chronos-GFP (green) in the ATN, followed by photostimulation combined with patch recording from a presubicular layer 3 cell. F, Amplitudes of oEPSCs evoked in layer 3 cells by blue (0.5 ms, 0.25 mW) or red light (2 ms, 0.6 mW) stimuli. G, Representative responses to light stimulation of ATN inputs. H, Injection of AAV5-Chrimson-tdTomato (red) in the RSC, followed by photostimulation combined with patch recording from a presubicular layer 3 cell. I, Amplitudes of oEPSCs evoked in layer 3 cells by blue (0.5 ms, 0.25 mW) or red light (2 ms, 0.6 mW) stimuli. J, Representative responses to light stimulation of RSC inputs K, Values and statistical analysis for different stimulation configurations (Panels F and I).

In order to quantify substrates of these innervations, we counted numbers of labeled ATN or RSC terminals located at less than 1  $\mu\text{m}$  from dendrites of biocytin filled layer 3 pyramidal cells ( $n = 6$ ; **Figure 3G**). The distribution of these potential contact sites on dendritic trees of layer 3 neurons was diverse. Some neurons exhibited highly segregated domains receiving either ATN or RSC inputs. For instance, cell 1 had a high proportion of clustered RSC putative synapses on its apical tuft while basilar dendrites were mostly surrounded by potential thalamic terminals. In cell 2, RSC terminals were clustered close to apical dendrites and also together with ATN terminals close to basilar dendrites. In contrast, many ATN terminals were located closer to apical dendrites of cell 3. Segregation of cortical RSC inputs to apical dendrites and thalamic ATN inputs to basal dendrites might favor supralinear EPSP summation. However, afferent terminal distributions differed and we found no clear pattern of sub- or supralinear summation (cell 1, supralinear summation (Dual/summed EPSP, 1.39); cell 2, roughly linear (0.91); cell 3, sublinear (0.69), **Figure 3G, H**). Furthermore, we noted certain potential contacts were located with separations less than 20  $\mu\text{m}$  on the same dendrite (**Figure 3I**). Such proximity might underly local dendritic computations and supra-linearities such as NMDA-mediated dendritic spikes which are suggested to contribute to non-linear summation of synaptic events (Larkum et al., 1999, 2007, 2009).

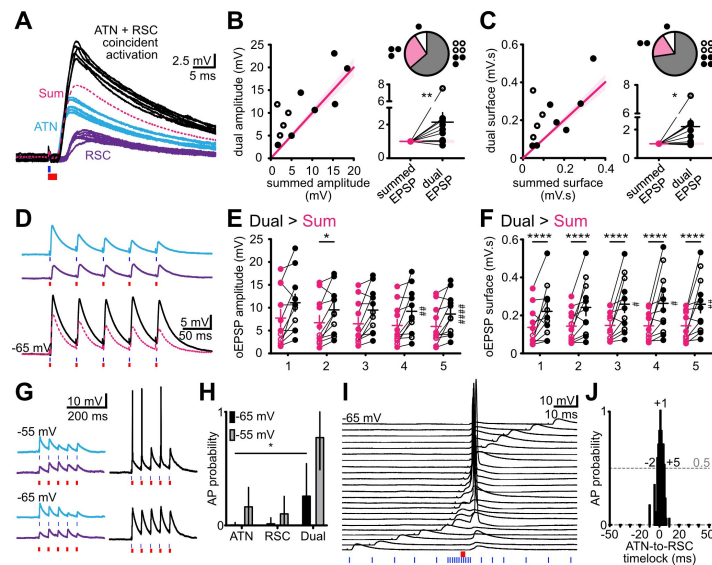
## Supralinear integration and amplification of ATN and RSC excitatory postsynaptic potentials

We explored interactions between excitatory inputs to layer 3 neurons by stimulating ATN and RSC axons separately with blue or red light. Excitatory synaptic potentials evoked by ATN or RSC axon stimulation, were compared to responses elicited when both sets of axons were activated at short time intervals (dual; **Figure 4A**). The mean amplitude of dual oEPSPs (synchronous light onset, **Figure 4A-C**) was significantly larger than calculated summed amplitudes of single oEPSPs ( $n = 11$  cells, dual/summed amplitude  $2.06 \pm 0.55$  mV,  $p = 0.0068$ , Wilcoxon test). Charge transfer, quantified as the surface under the dual oEPSP, was greater than the summed values for single oEPSPs (Dual/summed surface  $2.21 \pm 0.54$  mV.s,  $n = 11$ ,  $p = 0.0098$ , Wilcoxon test). These mean values cover variability between individual neurons. Summation of responses to ATN and RSC axon stimuli was supralinear in 7 out of 11 cells tested, linear for 3 cells and sublinear for 1 cell.

Dynamic responses to repetitive stimulations were also transformed by dual photostimulation. Amplitudes of 5 dual oEPSPs elicited at 20 Hz were higher than for a linear summation (Amplitudes in mV, Summed vs. Dual: Stim 1,  $7.7 \pm 1.9$  vs.  $11.1 \pm 1.9$ ; Stim 2,  $6.7 \pm 1.6$  vs.  $9.5 \pm 1.5$ ; Stim 3,  $6.5 \pm 1.4$  vs.  $9.5 \pm 1.5$ ; Stim 4,  $6.1 \pm 1.3$  vs.  $9.2 \pm 1.5$ ; Stim 5,  $5.9 \pm 1.3$  vs.  $8.6 \pm 1.4$ ,  $p = 0.0117$ , Two-way ANOVA; **Figure 4E**) and charge transfer, quantified as the surface under the 5 oEPSPs, also typically summed supralinearly (20 Hz, Surface in mV.s, Summed vs. Dual: Stim 1,  $136.9 \pm 30.6$  vs.  $221.2 \pm 41.5$ ; Stim 2,  $143.3 \pm 25.7$  vs.  $242 \pm 43.9$ ; Stim 3,  $147.3 \pm 22.5$  vs.  $259.2 \pm 44$ ; Stim 4,  $145.1 \pm 22$  vs.  $263.5 \pm 49.2$ ; Stim 5,  $145.6 \pm 23.2$  vs.  $258.7 \pm 45.2$ ,  $p = 0.0128$ , Two-way ANOVA; **Figure 4F**). Trains of dual ATN and RSC fiber stimulations also showed increased oEPSP integrals, compared to the stimulation of either one set of fibers alone, indicating stronger excitation over time (**Figure 4E, F**). Taken together, these data show supralinear summation of ATN and RSC inputs in most single layer 3 pyramidal cells, and also a facilitation of dynamic responses to repetitive inputs at 20 Hz.

We next asked how efficiently action potentials were induced by combined ATN and RSC inputs. Firing probability of layer 3 cells was higher for dual than for single input stimulation at a membrane potential of -65 mV (ATN 0.008 vs Dual 0.268,  $n = 5$ ,  $p = 0.0216$ , RSC 0.024 vs Dual 0.268,  $n = 5$ ,  $p = 0.1195$ , Friedman's and Dunn's tests) and higher still at -55 mV (**Figure 4G, H**). Firing was induced over a narrow window of time delays between stimulation of the two inputs (**Figure 4I, J**). It occurred for delays of -2 to +5 ms (RSC preceding ATN stimulus) reaching a maximum at a delay of +1 ms.





**Figure 4.**

### Supralinear summation of EPSPs and action potential firing following photostimulation of ATN and RSC axons.

A, Optical EPSPs in layer 3 neurons in response to blue light activation of ATN axons (cerulean traces), red light activation of RSC axons (purple traces). Supralinear summation of EPSPs following coincident activation of both ATN and RSC axons (black traces). The pink broken line indicates the calculated linear sum of oEPSPs evoked by stimulation of either ATN or RSC axons. Records from cell 1, [Fig 3D](#).

B, Amplitudes of dual ATN and RSC oEPSPs plotted as a function of the sum of separate ATN and RSC stimulations (*left*). Each circle is a cell ( $n = 11$  cells/4 mice), Pink line ( $\pm 10\%$ ) indicates linearity. Pie charts give the number of tested layer 3 neurons with supralinear (grey), linear (pink) or sublinear (white) summation. oEPSPs normalized to linear sum (*bottom right*). Solid circle, Chronos in ATN/Chrimson in RSC, empty circle, Chronos in RSC/Chrimson in ATN. \*  $p < 0.05$ , \*\*  $p < 0.01$ , from Wilcoxon test.

C, As in B, for dual ATN and RSC oEPSP integrals.

D, oEPSPs induced by 20 Hz stimulation of ATN (cerulean), RSC (purple) or both (black). The pink broken line indicates the calculated linear sum of oEPSPs evoked by separate ATN and RSC photostimulation.

E, Amplitudes of dual oEPSPs compared to those of the sum of ATN and RSC oEPSPs show that supralinearity increases across 5 stimulations. Two-way ANOVA,  $p = 0.0117$ . \*  $p < 0.05$ , \*\*\*\*  $p < 0.0001$ , Šidák's multiple comparison test. #  $p < 0.05$ , ##  $p < 0.01$ , Friedman's and *post-hoc* Dunn's test.

F, As in E, for dual ATN and RSC oEPSP integrals. Two-way ANOVA,  $p = 0.0128$ .

G, Excitatory postsynaptic responses to photostimulation of ATN (blue light, cerulean traces) or RSC axons (red light, purple traces) or both (blue and red light, black traces) at resting membrane potential (-65 mV) or at a depolarized holding potential (-55 mV). Synaptic excitation led to action potentials when dual ATN and RSC stimuli reached firing threshold.

H, Action potential (AP) probability for either or both stimuli at -65 and -55 mV. Data are presented as mean  $\pm$  SEM.

I, Action potentials were induced in presubicular layer 3 neurons by near coincident activation of ATN axons (red light) and RSC axons (blue light). Time delays varied from -50 to +50 ms.

J, Firing probability was maximal for short delays between -2 to +5 ms (RSC preceding ATN).



Supralinear summed responses of excitatory synaptic events may be mediated via the activation of NMDA receptors or voltage dependent intrinsic currents that amplify EPSPs (Fricker and Miles, 2000), especially if synaptic inhibition is reduced. We assessed these mechanisms in records made from layer 3 pyramidal neurons using a Cesium-based internal solution to favor cationic amplifying currents, and also containing QX314 to suppress action potentials (Figure 5). In 3 cells (from 2 mice) photostimulation of ATN fibers evoked oEPSPs of amplitude  $10.9 \pm 2.0$  mV and RSC fiber stimulation induced events of  $13.7 \pm 2.5$  mV. The response dynamics to 5 light pulses at 20 Hz were moderately facilitating (2nd vs 1st amplitude ratio: ATN 1.39, RSC 1.17; 5th vs 1st: ATN 1.18, RSC 1.20;  $n = 3$ ). Coincident activation of ATN and RSC afferents induced supralinear oEPSP summation. Furthermore, repetitive stimulation generated large all-or-none depolarizations on the second or third stimulus at 20 Hz (2nd vs 1st amplitude ratio: 1.52; 5th vs 1st: 2.90;  $n = 3$  cells, Figure 5A). It may be mediated by VGCC or a QX-314 resistant  $\text{Na}^+$  inward current (Fricker et al., 2009). This component appeared with higher probability and shorter latency in the presence of GABA<sub>A</sub> receptor blocker gabazine (Figure 5B). The NMDA receptor antagonist APV largely abolished the dual EPSP amplification initially (Figure 5C, black trace; 2nd vs 1st integral ratio: 1.22; 5th vs 1st: 1.32), although amplification was partially restored by increasing RSC stimulus intensity (pale pink trace; “amplification” refers to the shape of the plateau-like prolongation of the peak, most pronounced on the second EPSP, indicated with an arrow). NMDA receptor activation thus assists depolarization towards the threshold of a voltage-dependent process contributing to supra-linear EPSP summation, but is not the only charge carrier involved.

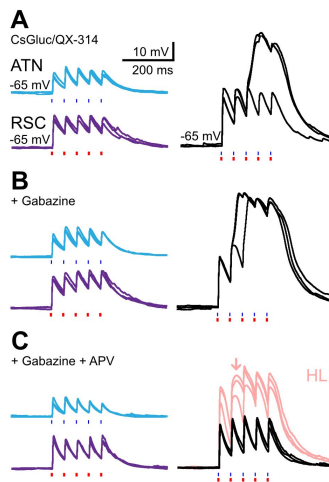
## Cholinergic modulation and recruitment of presubicular layer 4 neurons by ATN and RSC afferents

Presubicular layer 4 neurons are intrinsic bursting pyramidal neurons that project to the lateral mammillary nucleus (Huang et al., 2017). This pathway is critical for the coordinated stable landmark control of HD cells in the thalamus and throughout the HD cell circuit (Yoder et al., 2015; Yoder et al., 2017). To investigate the input connectivity of layer 4 principal cells, we recorded responses of these neurons to stimulation of ATN and RSC afferents.

Layer 4 neurons labeled by retrograde tracers injected in the lateral mammillary nucleus were located in the *lamina dissecans* of the presubiculum, below layer 3, where thalamic axons ramify (Figure 6A–D). At the proximal end of the presubiculum, and in its continuity (Ishihara and Fukuda 2016), some subicular neurons projecting to the medial portion of the mammillary bodies are also labeled (Figure 6B). The apical dendrites of layer 4 presubicular neurons extended towards presubicular layer 1 as previously described (Huang et al., 2017), but tended to circumvent layer 3 and avoid thalamic afferents, by swerving towards the subiculum. Apical dendrites of some neurons crossed layer 3 obliquely, while others avoided thalamic afferents in layer 3 (Figure 6C, D, Figure 6—figure supplement 1).

Layer 4 neurons had a more depolarized resting membrane potential, lower input resistance and time constant than layer 3 neurons (Figure 6—figure supplement 2 compares active and passive properties). A characteristic voltage sag in responses to hyperpolarizing steps, indicated the presence of an  $I_h$  current. Layer 4 neurons discharged bursts of two or three action potentials at the onset of a depolarizing step current injection and also after the offset of hyperpolarizing steps (Figure 6E). These bursts were abolished by the T-type  $\text{Ca}^{2+}$  channel blocker TTA-P2 (Figure 6F).

We next recorded responses of layer 4 neurons to optical activation of ATN or RSC afferents together with effects of the same stimuli on layer 3 pyramidal cells (Figure 6G). Overall, latencies of oEPSCs in layer 4 neurons were longer than for layer 3 (ATN layer 4,  $6.2 \pm 0.6$  ms,  $n = 8$ , vs. layer 3,  $2.4 \pm 0.2$  ms,  $n = 24$ ; RSC layer 4,  $5.6 \pm 0.4$  ms,  $n = 9$ , vs. layer 3,  $2.9 \pm 0.2$  ms,  $n = 27$ ;



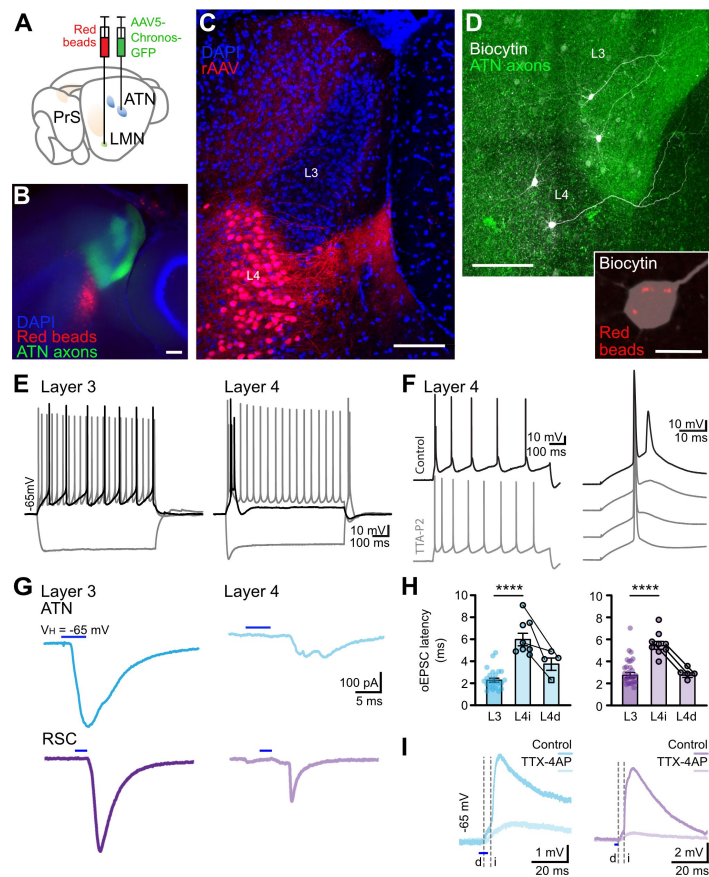
**Figure 5.**

### EPSP amplifications in layer 3 pyramidal neurons.

A, Optical EPSPs evoked in layer 3 pyramidal neurons following the stimulation of ATN axons (blue light, cerulean traces), RSC axons (red light, purple traces) or both (blue and red light, black traces). Recording pipette contained a cesium gluconate based internal solution and the  $\text{Na}^+$  channel blocker QX-314. A large all-or-none EPSP amplification occurred for dual stimuli at 20 Hz, on some trials.

B, In the presence of the  $\text{GABA}_A$  receptor antagonist gabazine (10  $\mu\text{M}$ ), dual EPSP amplification occurred earlier in the train.

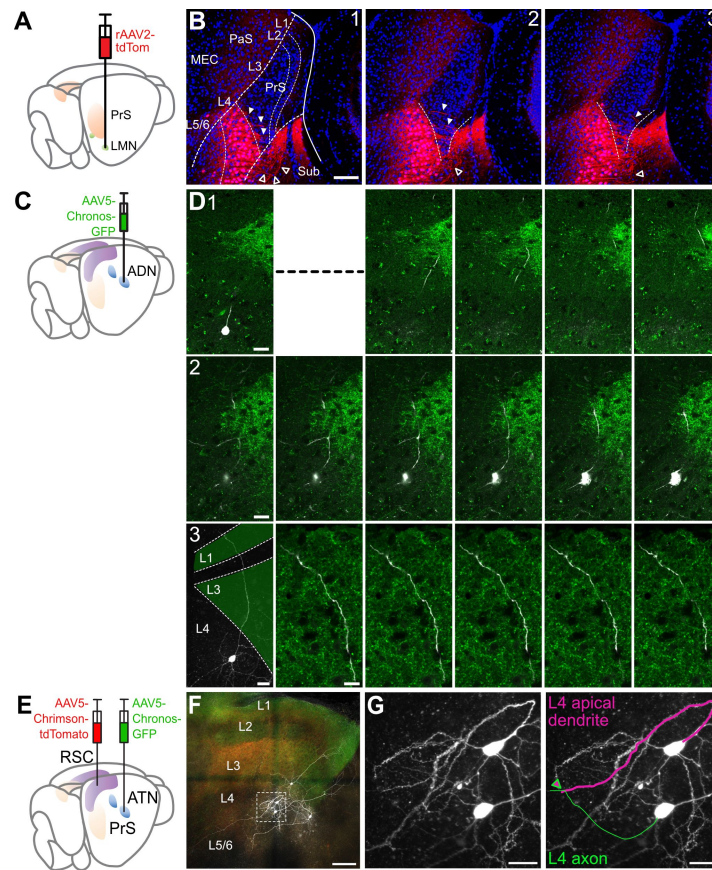
C, The additional presence of the NMDA receptor antagonist APV (100  $\mu\text{M}$ ) abolished dual EPSP amplification (black trace). EPSP amplification was partially restored by increasing red light intensity x2 (pale pink traces).



**Figure 6.**

**Presubicular LMN-projecting layer 4 neurons avoid thalamo-recipient layer 3 and receive little direct input from ATN and RSC.**

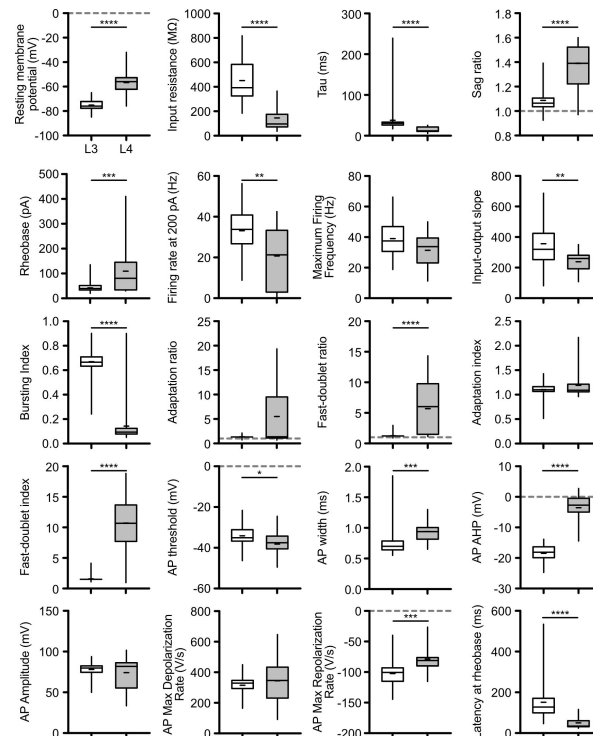
A, Expression of Chronos-GFP in ATN and retrograde labeling of neurons that target LMN.  
 B, Thalamic axons (green) in superficial layers 1 and 3 of presubiculum. Retrobeads label cell bodies of presubicular layer 4 cells (red).  
 C, Retrograde rAAV2-tdTomato label cell bodies and dendrites of layer 4 LMN projecting neurons (red). Apical dendrites of layer 4 pyramidal neurons avoid layer 3 where thalamic axons ramify.  
 D, Presubicular slice containing two layer 3 and two layer 4 neurons filled with biocytin (white) and GFP expressing thalamic axons (green). Scale bar 100  $\mu$ m. Inset, retrobeads (red) in the soma of a biocytin filled LMN-projecting layer 4 neuron. Scale bar 10  $\mu$ m.  
 E, Layer 3 neurons are regular spiking and layer 4 neurons are burst firing, initially and at rebound, in response to current injection. Black trace, rheobase.  
 F, T-type  $\text{Ca}^{2+}$  channel blocker TTA-P2 (1  $\mu$ M) suppressed burst firing in presubicular layer 4 neurons, while single action potentials were preserved.  
 G, Representative oEPSCs in layer 3 (left) and layer 4 (right) pyramidal cells, in response to stimulation of ATN (cerulean) or RSC (purple) inputs.  
 H, oEPSC latencies in layer 3 and layer 4 cells, for ATN inputs (left, cerulean), or RSC inputs (right, purple). Each dot is a cell. Same layer 4 cells are indicated by connecting lines, to show the difference in latency for direct and indirect synaptic responses.  
 I, oEPSCs in layer 4 neurons in response to stimulation of ATN (cerulean) or RSC (purple) inputs in control and in the presence of TTX (1  $\mu$ M) and 4-AP (100  $\mu$ M). Dashed lines indicate the timing of the large disynaptic component of the responses (i, indirect), and the small monosynaptic response (d, direct), isolated in TTX-4AP.  
 See also Figure 6—figure supplements 1 and 2.



**Figure 6—figure supplement 1.**

### Apical dendrites of presubicular layer 4 neurons avoid the thalamorecipient layer 3.

A, Retrograde AAV2retro-tdTom was injected in LMN. B, PrS L4 neuron cell bodies and dendrites were labeled. Scale bar, 50  $\mu$ m. Full arrowheads, L4 neurons apical dendrites crossing L3; Empty arrowheads, L4 neurons apical dendrites skirting L3. C, Anterograde AAV5-Chrimson-GFP was injected in ATN to label thalamic axons, delimiting PrS layer 3. D, Examples of biocytin filled layer 4 presubicular neurons. (1) The apical dendrite of this neuron avoided the densely packed ATN afferents and it didn't show responses to thalamic stimulations. (2) This neuron's apical dendrite crossed through a short stretch of layer 3 containing ATN afferents, and it responded with low amplitude indirect events. (3) The apical dendrite of this neuron crossed layer 3 and the zone containing ATN afferents, and was directly recruited. Scale bar 10 $\mu$ m. E, AAV-Chrimson-GFP injected in ATN, and AAV-Chrimson-tdTomato injected in RSC. F, Image from **Figure 7** panel D. Scale bar 100 $\mu$ m. G, (Left) Enlarged view of dotted area in F, with two labelled L4 neurons. The apical dendrite of top neuron makes a U-turn to ramify in deep layers and seems to cross path with the axon of the bottom neuron, potentially forming a synaptic contact. (Right) The dendrite of the top neuron and axon of bottom neuron are overlaid with pink and green lines respectively. Scale bar 20 $\mu$ m.



**Figure 6—figure supplement 2.**

### Electrophysiological passive and active intrinsic properties of layer 3 vs. layer 4 neurons.

Data for layer 3 are summarized in **Figure 2—figure supplement 1** and are shown here for comparison. Statistical differences are indicated as levels of p values (\*  $p < 0.05$ , \*\*  $p < 0.01$ , \*\*\*  $p < 0.001$ , \*\*\*\*  $p < 0.0001$ ) obtained from Mann-Whitney tests between layer 4 ( $n = 19$ ) and layer 3 ( $n = 33$ ) for each parameter. Box plots indicate maximum and minimum values, quartiles, full line is the median, short line is the mean.

**Figure 6H** [↗](#)), indicating possible polysynaptic excitation of layer 4 neurons. Bath application of TTX-4AP did not abolish oEPSPs entirely, leaving a low amplitude component with short, potentially monosynaptic latencies (latency ATN  $3.9 \pm 0.6$  ms, RSC  $2.9 \pm 0.2$  ms,  $n = 5$ ; **Figure 6I** [↗](#)).

Comparison of the timing of synaptic events and firing (**Fig 7A-C** [↗](#)) showed that oEPSP onset in layer 4 neurons occurred after firing in the layer 3 neuron, following ATN afferent stimuli, in 4 out of 5 cell pairs. We also observed this sequence when RSC input was activated, in one tested pair. Depolarization of the layer 3 cell via the patch pipette, initiated firing, but EPSPs were not elicited in any simultaneously recorded layer 4 neuron (0 out of six cell pairs tested). These data suggest that excitation of layer 3 cells by ATN and likely by RSC afferents is transmitted to layer 4 neurons, even in the absence of direct evidence for mono-synaptic coupling between cell pairs.

In records from pairs of layer 3 and 4 neurons ( $n = 3$ ; **Figure 7D** [↗](#), E) layer 3 cells responded with precisely timed action potentials to low intensity stimulation of ATN afferents, while only very small oEPSPs were initiated in layer 4 neurons. Increasing the excitatory drive by stimulating RSC afferents elicited larger oEPSPs more reliably in layer 4 neurons. Higher intensity stimulation of both ATN and RSC axons, could evoke bursts of action potentials, with the activation of an underlying  $Ca^{2+}$  current in layer 4 cells (**Figure 7E** [↗](#)). We hypothesize that increasing the activity of layer 3 neurons by strong (and non-specific) stimulation of ATN and RSC afferents is needed to induce discharges in layer 4 cells. We should also note that strong photostimulation of a single set of afferent fibers could initiate discharges in layer 4 pyramidal cells, as was shown in records from mice where only one afferent brain area, ATN or RSC, expressed an opsin (**Figure 7—figure supplement 1** [↗](#)).

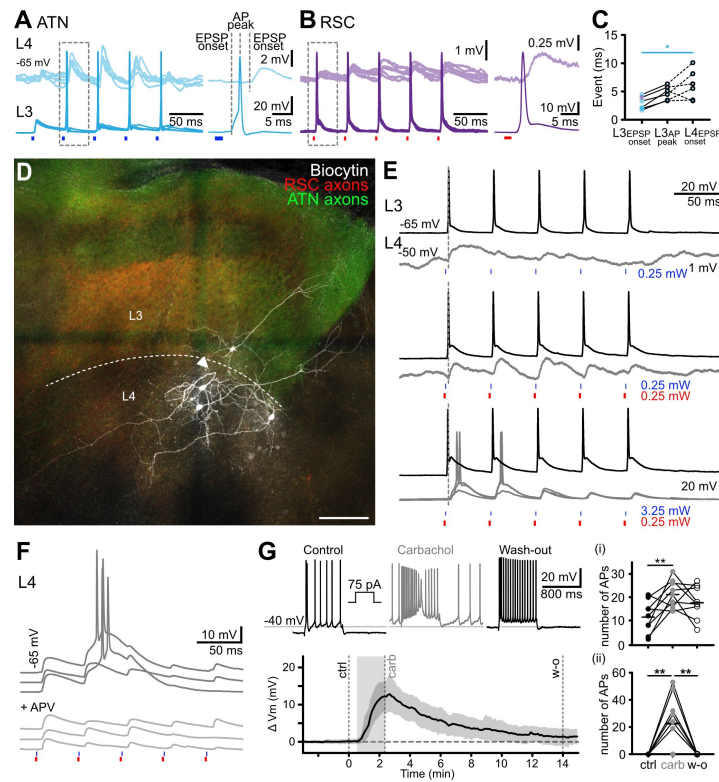
Overall, it seems probable that layer 3 neurons relay activity projected by ATN and RSC inputs onto layer 4 pyramidal neurons. However, evidence on this point could be improved. In  $n = 9$  paired recordings, we did not detect functional synapses between layer 3 and layer 4 neurons. Anatomically, putative synaptic contacts between filled axons of layer 3 and dendrites of layer 4 cells, were not evident ( $n = 5$  filled layer 4 cells; see also Peng et al., 2017 [↗](#)). We did find a possible synaptic connection between two neighboring L4 neurons in one case (**Figure 6—figure supplement 1D** [↗](#)), with a close apposition between axon and dendrite. Recurrent excitation may promote bursting in a positive feedback loop in layer 4 neurons, and NMDA receptor related EPSP amplification favors burst firing (**Figure 7F** [↗](#)).

Neuromodulatory factors may help drive layer 4 cells to fire action potentials. As Presubiculum is rich in acetylcholinesterase (Slomianka and Geneser 1991 [↗](#)), we examined excitability changes in the presence of a broad cholinergic agonist. The application of Carbachol ( $10 \mu\text{M}$ ; **Figure 7G** [↗](#)) led to a depolarization of the recorded layer 4 neurons' membrane potential by  $13 \pm 4$  mV, with increased action potential firing during a positive step current injection, and from the baseline (**Figure 7Gi,ii** [↗](#)).

## Discussion

Understanding the anchoring of HD signals to environmental landmarks provides a useful framework to ask how external information is integrated to revise and improve brain representations. Here we show that thalamic ATN afferents signaling HD data and cortical RSC projections that carry landmark information converge on layer 3 pyramidal cells of the dorsal presubiculum. Independent optogenetic stimulation of these two glutamatergic afferents in slices was used to define mechanisms of their integration in the presubiculum. Most layer 3 cells were innervated by both ATN and RSC fibers, sometimes close on the same dendritic branches, sometimes on different dendrites. Nearly coincident EPSPs (2-5 ms) induced independently by ATN and RSC afferents, evoked non-linear membrane responses to trigger layer 3 cell firing, transmitted to the MEC. In a second processing step, layer 3 neurons also excite layer 4 pyramidal





**Figure 7.**

### Cross-laminar activation of LMN-projecting layer 4 bursting neurons.

A, Simultaneous records of a layer 4 and a layer 3 neuron during photostimulation of ATN afferents. EPSP onset was delayed in the layer 4 neuron. Right panel, expanded view of the boxed area.

B, Simultaneous records of a layer 4 and a layer 3 neuron during photostimulation of RSC afferents. As in A, layer 4 neurons responded with a delay. Right panel, expanded view of the boxed area.

C, Latencies of synaptic activation indicated by the dotted lines in A, B (ATN stimulation,  $n = 4$  cells; RSC stimulation,  $n = 1$ ). EPSP onset and AP peak from layer 3 neurons ( $n = 5$ ). Dotted lines link to the EPSP onset in simultaneously recorded layer 4 neurons ( $n = 5$ ).  $p < 0.05$ , Kruskal-Wallis multiple comparison test.

D, Biocytin labeled layer 3 and layer 4 pyramidal neurons, in a presubicular slice containing thalamic (green) and retrosplenial (red) axons. The apical dendrite of one layer 4 neuron makes a U-turn (arrowhead), away from layer 3, where the thalamic axons ramify. The neighboring layer 4 neuron's apical dendrite crosses the thalamo-recipient layer 3 for a short distance before arborizing outside of ATN targeted area, towards the subiculum, on the right. Another biocytin filled layer 4 neuron's dendrites extend toward the deep layers. Scale bar,  $100\mu\text{m}$ .

E, Simultaneous records from a layer 3 and a layer 4 cell to ATN input stimulation (top,  $0.25\text{ mW}$ , blue light), ATN and RSC input stimulation (middle,  $0.25\text{ mW}$  blue and  $0.25\text{ mW}$  red light; light intensities compatible with independent photostimulation) or non-specific ATN and RSC input stimulation (bottom,  $3.25\text{ mW}$  blue and  $0.25\text{ mW}$  red light). ATN fibers expressed Chronos-GFP (green) and RSC fibers expressed Chrimson-tdTomato (red).

F, Top, oEPSPs and bursts of action potentials in a layer 4 neuron, evoked by dual wavelength stimulation of ATN and RSC afferents at  $20\text{ Hz}$ . Amplifications of dual oEPSPs led to firing. Bottom, the NMDA receptor antagonist APV ( $100\mu\text{M}$ ) reduced EPSP amplification and prevented action potential firing.

G, Layer 4 bursting neurons are sensitive to the acetylcholine receptor agonist carbachol ( $10\mu\text{M}$ ). Action potential firing in response to step current injections in control (black), in the presence of carbachol (grey), and after wash-out (black). Bottom graph, membrane potential depolarization during a 2-minute carbachol application. The number of action potentials increased during the depolarizing steps (i) and on the baseline (ii).

See also **Figure 7—figure supplement 1**.

Figure 7—figure supplement 1.

### Regular firing layer 3 vs. intrinsically bursting layer 4 neurons responded to high-intensity light stimulations of ATN or RSC afferents.

Layer 3 neurons respond with single spikes following each light stimulation while layer 4 neurons respond with bursts of action potentials at the beginning of the stimulation train.

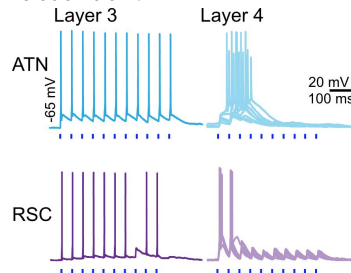
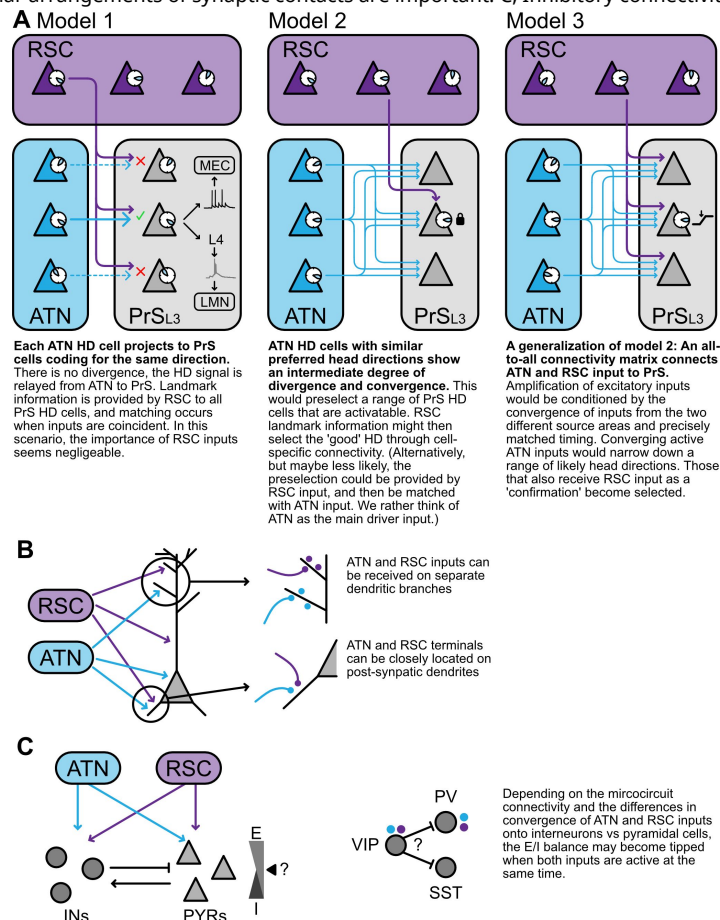


Figure 7—figure supplement 2.

### Schematic models for landmark anchoring of HD signals.

A, different scenarios with varying degrees of divergence and convergence of the two input sources. We note that matching directional with visual landmark information does not necessarily require synaptic plasticity, and might rely on temporal coincidence. B, subcellular arrangements of synaptic contacts are important. C, Inhibitory connectivity is important.



cells which receive little direct ATN and RSC innervation. These burst firing neurons may project a distinct, di-synaptically mediated, visually-updated HD signal to the LMN. Our data thus suggest that the dorsal presubiculum integrates HD and landmark signals producing two distinct output signals which are transmitted to different regions. Cholinergic modulation may facilitate responses to salient stimuli, for flexible anchoring to landmarks.

## Anatomical convergence of ATN and RSC projections in the dorsal Presubiculum

Retrograde tracing in this study confirmed strong projections to the presubiculum from the ATN and the RSC to the presubiculum (**Figure 1** [↗](#)). Anterior thalamic fibers (Van Groen et Wyss 1990c; 1990a; 1990b; 1992; Shibata et Honda 2012; Vogt et Miller 1983) ramify in and delimit the anatomical borders of the presubiculum (Liu et al., 2021 [↗](#); Simonnet et al., 2017 [↗](#)). HD signals from HD cells of the anterior thalamus project to the presubiculum (Goodridge and Taube, 1997 [↗](#)) where they form synapses directly with layer 3 pyramidal cells (Nassar et al., 2018 [↗](#)). The retrosplenial cortex was the most strongly labeled cortical region innervating the presubiculum. We detected retrogradely transported beads in cells of layers 2 and 5 of dysgranular RSC, and layer 5 of the granular RSC across its antero-posterior axis (Sugar and Witter, 2016 [↗](#)).

We examined the anatomy and physiology of these two afferent systems to ask how visual landmark signals from the RSC are combined with HD signals relayed via the ATN. Anterograde fiber tracing, with an AAV5 construct expressing Chronos-GFP, confirmed that axons from both regions project to superficial layers of the dorsal presubiculum. No RSC projections were found in ventral PrS for injections in rostral RSC, as previously noted (Jones and Witter 2007 [↗](#); Kononenko and Witter 2012 [↗](#)). Both RSC and ATN innervated superficial presubicular layers 1 and 3, while sparing layer 2 containing somata of calbindin positive neurons (**Figure 1K** [↗](#) and cf. Balsamo et al. 2022 [↗](#)). We detected microzones containing a high density of ATN-positive fibers, but no RSC fibers, in upper layer 3. It remains to be shown how these anatomical inhomogeneities may translate to functional modularity.

## Pathway specific functional connectivity onto PrS layer 3 neurons

Photostimulation of GFP-Chronos expressing axons let us compare synaptic events initiated in layer 3 pyramidal cells by fibers from the ATN or the RSC in dorsal presubiculum slices (**Figure 2** [↗](#)). Both ATN and RSC afferents formed mono-synaptic glutamatergic connections with components mediated by NMDA and AMPA receptors. The amplitudes of synaptic currents varied between neurons, with an overall similar amplitude distribution for ATN and RSC stimulation. This may be due to variable expression levels (Hooks et al. 2015 [↗](#)), or may imply variations in coupling weights across cells. ATN and RSC fiber mediated EPSCs depressed during repetitive activation at 20 Hz. EPSPs induced firing early during repetitive stimulation. RSC inputs tended to produce more sustained EPSP dynamics and slower rise times than ATN afferents for low intensity stimulation. ATN synapses with presubicular pyramidal cells resemble those made by thalamic afferents in the somatosensory cortex - high release with depressing dynamics, (Gil et al., 1999 [↗](#)) possibly due to presynaptic expression of VGLut2 (Liu et al., 2021 [↗](#)).

Dual wavelength optogenetic stimulation was used for independent activation of intermingled afferents expressing blue-light sensitive Chronos in ATN fibers and red-shifted Chrimson in RSC fibers (Klapoetke et al., 2014 [↗](#)). Precautions were taken to avoid cross-stimulation since all channelrhodopsin variants are somewhat sensitive to blue light. With the fast, sensitive opsin Chronos expressed in ATN fibers, synaptic events were induced by very short (0.5 ms), low intensity, 0.25 mW, stimuli. For RSC fibers expressing Chrimson, using light stimuli of duration 2 ms, adjusting intensity up to 2 mW initiated synaptic events of comparable amplitude. Calibration experiments (**Figure 3—figure supplement 1** [↗](#)) provided strong evidence for the independence

of responses. While Chrimson has slower dynamics than Chronos (Klapoetke et al., 2014), synaptic events induced by stimulating either opsin were similar, as we showed by swapping the two opsin variants injected in ATN and RSC fibers respectively (**Figure 4**).

This dual opsin approach permitted independent stimulation with blue and red light pulses. Most (76%) recorded layer 3 neurons generated synaptic events in response to stimulation of both ATN and RSC fibers providing a substrate for integration of landmark information from the RSC with thalamic HD signals. Layer 3 cell firing was most effectively triggered by nearly coincident inputs (-2 to +5 ms separation, **Figure 4I, J**). This is of interest since HD representations in the AD and RSC are highly coherent (Fallahnezhad et al., 2023), with short intervals between spikes in these two regions (< 5ms; van der Goes et al. 2022). Converging inputs from thalamic and retrosplenial axons can therefore excite common postsynaptic presubicular layer 3 neurons with very short delays, such that coincidence detection by these cells will tend to enhance HD signals. Response dynamics to combined stimulation of both inputs at 20 Hz were maintained or facilitating, in contrast to the depressing dynamics of repetitive stimulation of one set of afferent fibers. Combined and temporally precise inputs from ATN and RSC may thus help maintain HD signaling during immobility. Our findings may well underlie the recent in vivo observation by Siegenthaler et al (2024) that visual objects refine HD coding, and that presubicular HD cells whose preferred firing direction corresponds to a visual landmark respond with higher firing rates.

## Nonlinear signal integration in layer 3 thalamo-recipient neurons and physiological significance for HD-to-landmark anchoring

Convergence of ATN and RSC axons onto single layer 3 pyramidal cells provides the anatomical basis of synaptic integration. Putative synaptic contacts from both afferent fiber systems were found on the basal dendrites of pyramidal cells (**Figure 3**), sometimes on the same branch. Photostimulation centered on the soma of recorded neurons predominantly activated synapses on basal dendrites. Supralinear summation could result from local spike-generating mechanisms if the activated synapses were located on a same dendritic branch (Makarov et al., 2023; Poirazi and Papoutsis, 2020; Polsky et al., 2004). Clustered synapses could also guide the formation of new spines and synapses during integration of landmark information in the HD signal. Learning might bind new inputs into functional synaptic clusters (Hedrick et al., 2022). Our small sample of layer 3 pyramidal cells suggest that both ATN and RSC axons target basal dendrites, and more RSC than ATN axons contact apical dendrites (**Figure 3**). Tests on the effects of precise, near coincident activation of basal and apical synapses could be revealing but were not technically possible in this study. Distinct dendritic inputs to layer 3 cells may improve spatial perception (Takahashi et al., 2016) enhancing HD signal quality by integration with landmark information.

Our data show that EPSPs elicited by nearly coincident ATN and RSC inputs which exceed a threshold are amplified by NMDA receptor activation and voltage gated inward dendritic currents (**Figure 5**; Fricker et al. 2009). From a small sample of cells it appears that EPSP amplification may be facilitated by a reduction in synaptic inhibition (n = 3; **Figure 5**), indicating that disinhibition may be permissive for supralinearity and gate firing by dynamic modulation of the balance between inhibition and excitation (Milstein et al., 2015). VIP expressing interneurons, which are excited by cholinergic modulation could provide such disinhibition of the presubicular microcircuit (Porter et al., 1999; Slomianka and Geneser, 1991). The excitation-inhibition balance in pyramidal cells may become tipped towards excitation in the case of coincident, co-tuned thalamic and retrosplenial input. A ‘when-to-learn’ signal for HD updating in the presubiculum might function analogously to the promotion by dopamine of associations between sensory cues and head direction cells in the fly (Fisher et al., 2022). It is however unclear whether LTP type learning takes place in the mammalian head direction circuit. Presubicular layer 3 cells do not express the GluR1 subunit of AMPA receptors (Martin et al 1993; Ishihara et al., 2016) that is critical for LTP expression (Boehm et al., 2006). The absence of GluR1 might

indicate that the thalamo-presubicular synapses in layer 3 function without classical long-term synaptic plasticity. Matching directional with visual landmark information based on temporal coincidence may be sufficient. Algorithms for dynamic control of cognitive maps without synaptic plasticity (Whittington et al 2025 [Neuron](#)) propose that working memory is stored in neural attractor activity, and updated by recurrent connections, which might generalize to the HD system.

Layer 3 pyramidal cells may be described as multi-compartmental computational devices (Häusser and Mel, 2003 ; Mel, 1993 ; Poirazi et al., 2003 ; Spruston, 2008 ) which integrate HD and landmark information. ATN axons drive presubicular HD neurons (Peyrache et al., 2015 ) and RSC mixed selectivity neurons contribute visual landmark information and allocentric spatial references (Jacob et al., 2017 ; Mitchell et al., 2018 ; Vann et al., 2009 ). NMDA mediated dendritic spikes enhance tuning selectivity in visual cortex (Smith et al., 2013 ; Wilson et al., 2016 ) and barrel cortex (Lavzin et al., 2012 ). Dendritic events in Layer 3 PrS cells may enable binding of visual landmarks with HD tuning. It is tempting to speculate that nonlinear synaptic integration and inhibitory gating may be involved in flexibly updating the allocentric direction of HD cells based on the integration of visual landmark information to the current HD signal (Siegenthaler et al., 2024 ). In the primary sensory cortex, nonlinearities act to increase perceptual threshold of sensory information (Takahashi et al., 2016 ; 2020 ). The attractor network in the PrS could thus be either stabilized or flexibly reset to external spatial cues.

Alternative models of HD-to-landmark anchoring have focused on the RSC, which also treats HD and landmark signals (Page & Jeffrey, 2018; Yan, Burgess, Bicanski, 2021 ). Possibly the RSC registers a direction to a landmark rather than comparing it with the current HD (Sit & Goard, 2023 ). The integrated information would then reach PrS, which in contrast to RSC, is uniquely positioned to update the signal in the LMN (Yoder et al., 2011 ).

What is the neuronal and circuit mechanism for integrating the “right” set of landmarks to anchor the HD signal? In our slice work we are blind to the exact nature of the signal that is carried by ATN and RSC axons. Critical missing information to understand the logic of HD-to-landmark anchoring also includes the degree of divergence and convergence of connections from a single thalamic or retrosplenial neuron. Do HD tuned inputs from thalamus converge on similarly tuned HD neurons only? Is divergence greater for the retrosplenial inputs? If so, thalamic input might pre-select a range of HD neurons, and converging RSC input might narrow down the precise HD neurons that become active (**Figure 7—figure supplement 2** ). In the future, activity dependent labeling strategies might help to tie together information on the tuning of pre-synaptic neurons, and their convergence or divergence onto functionally defined postsynaptic target cells.

## Functional significance of two-layer processing preceding projection to LMN

Burst-firing layer 4 PrS cells project a distinct version of HD-landmark signals to the LMN. The dendrites of these LMN projecting neurons overlap little with direct ATN and RSC inputs granting partial isolation from direct excitation, and permitting a second level of integration of information from layer 3 neurons. Layer 3 axons may innervate layer 4 neurons basal dendrites, which may be driven to fire by sufficiently strong excitatory inputs from layer 3, especially if combined with cholinergic activation.

There are advantages to segregating the integration of converging input signals from the updating signal across layers. Segregation permits both a fast transmission of an integrated signal to the medial entorhinal cortex, and the conditional transmission of an updating signal mediated by burst firing to upstream HD circuit elements. The synaptic threshold implicit in this two-stage system permits a gated updating. Functionally layer 4 neurons are uniquely positioned to update the HD signal in the LMN with visual landmark information (Yoder et al., 2015 ). In this way, cell-type specific cholinergic facilitation may help idiothetic cue based navigation (Yoder et al.,



2017 [↗](#)). These cellular and synaptic circuit data support and expand the findings of Yoder et al. (2015 [↗](#) and 2017). They clarify how the thalamic HD signal integrated with visual landmark information is relayed to the grid cell system in the medial entorhinal cortex and to the lateral mammillary nuclei.

## Limitations

Burst firing may play a role for learning in hierarchical circuits (Naud et al., 2023 [↗](#); Payeur et al., 2021 [↗](#)). In the HD circuit, landmark information contained in bursts might reset or anchor the HD attractor in the LMN and beyond. The effects of burst firing signals transmitted to the LMN remain to be assessed. Potentially HD signals may be updated with visual signals at several sites including the PrS, but PrS uniquely provides a feed-back projection to the lateral mammillary nucleus. Further work on layer 3 to layer 4 transmission is warranted, and the link to spatial perception and behavioural updating needs to be strengthened.

## Methods

### Animals

Experiments were performed on male and female wild-type and transgenic (Ai14 reporter line - Jax n007914) C57Bl6 mice of age 35-70 days, housed on a 12-hours light/dark cycle with food and water available *ad libitum*. Animal care and use conformed to the European Community Council Directive (2010/63/EU) and French law (87/848). Our study was approved by the local ethics committee (CEEA - 34) and the French Ministry for Research 01025.02.

### Viral vectors and beads

Projecting neurons were labeled with retrograde fluorescent tracers (Retrobeads, Lumafluor, and AAV2retro-CAG-tdTomato, Addgene 59462P). Fluorescent beads were stored at 4°C before use. Channelrhodopsin expression was achieved by injecting Adeno-associated viral constructions. AAV5.Syn.Chronos-GFP.WPRE.bGH (AAV5-Chronos, Penn Vector Core, Addgene 59170P) was used to induce neuronal expression of the blue light induced channelrhodopsin Chronos fused to the GFP marker and under the control of the Synapsin promoter. AAV5.Syn.ChrimsonR-tdTomato.WPRE.bGH (AAV5-Chrimson, Penn Vector Core, Addgene 59171P) induced neuronal expression of the red light gated channelrhodopsin Chrimson fused to the tdTomato marker, under the control of the Synapsin promoter. Viral vectors were stored at -80°C before use.

### Stereotaxic surgery

Mice at ages of 4-5 weeks were anesthetized by intraperitoneal (i.p.) injection of a mixture of ketamine hydrochloride and xylazine (100 and 15 mg/kg respectively, in NaCl 0.9%). They were placed in a stereotaxic frame for injections. Fluorescent beads for retrograde tracing were injected (300-500 nl) into the PrS at the coordinates: -4.06 antero-posterior (AP), 2.00 medio-lateral (ML) and -2.15 mm dorso-ventral (DV) and into the LMN (-2.8 AP, 0.75 ML, -5.35 DV) with respect to the bregma. Note, due to the small size of LMN, some overflow to the medial mammillary nucleus (MMN) may have occurred.

Viral injections were performed unilaterally (Mathon et al., 2015 [↗](#); Richevaux et al., 2019 [↗](#)) at the coordinates -0.82 AP, 0.75 ML and -3.2 mm DV for the ADN, and at -2.1 to -2.15 AP, 0.65 ML and -0.65 mm DV for the RSC. Volumes of 200 to 250 nl were injected with a 10 µL Hamilton syringe equipped with 33ga needle over a time of 10 min. The needle was slowly removed after a delay of 10 min to avoid leakage from the injection site. Best expression of AAV5 serotypes was achieved after 3 to 4 weeks.



## Tissue fixation and slicing for retrograde tracing

Brains were removed for anatomy at 4 days after retrobead injection. Mice were anesthetized by i.p. injection of the ketamine/xylazine mixture. An intracardiac perfusion with 0.1M PBS was followed by perfusion with 4% paraformaldehyde. Brains were stored overnight in paraformaldehyde at 4°C and then washed in PBS. Coronal or horizontal sections were cut at 100  $\mu\text{m}$  with a vibratome and stored in sucrose at 4°C.

## Preparation of brain slices for physiology

Slices of the temporal lobe were prepared 3-4 weeks after injection of AAV5 viral constructions. Mice were anesthetized by i.p. injection of the ketamine/xylazine mixture. They were then perfused intracardially with a cutting solution containing (in mM): 125 NaCl, 25 sucrose, 2.5 KCl, 25  $\text{NaHCO}_3$ , 1.25  $\text{NaH}_2\text{PO}_4$ , 2.5 D-glucose, 0.1  $\text{CaCl}_2$ , 7  $\text{MgCl}_2$ , cooled to 4°C, and oxygenated with a 5%  $\text{CO}_2$  / 95%  $\text{O}_2$ . The brain was removed and a vibratome was used to cut horizontal slices at 300  $\mu\text{m}$  in the same solution. Slices were stored for 15 min at 34°C in an ACSF containing (in mM): 124 NaCl, 2.5 KCl, 26  $\text{NaHCO}_3$ , 1  $\text{NaH}_2\text{PO}_4$ , 2  $\text{CaCl}_2$ , 2  $\text{MgCl}_2$ , and 11 D-glucose, bubbled with 5%  $\text{CO}_2$  / 95%  $\text{O}_2$ . They were then kept in the same solution at room temperature until recording.

## Whole-cell patch-clamp recordings

Slices were transferred to a recording chamber perfused with oxygenated, warmed ( $\sim 32^\circ\text{C}$ ) ACSF mounted on an epifluorescence microscope. Patch-clamp records were made from neurons with borosilicate glass pipettes of external diameter 1.5 mm (Clark Capillary Glass, Harvard Apparatus) pulled with a Brown-Flaming electrode puller (Sutter Instruments). Electrodes, filled with a potassium-gluconate based solution containing (in mM): 135 K-gluconate, 1.2 KCl, 10 HEPES, 0.2 EGTA, 2  $\text{MgCl}_2$ , 4 MgATP, 0.4 Tris-GTP and 10  $\text{Na}_2$ -phosphocreatine, had a resistance of 4-8 M $\Omega$ . An alternative, cesium-gluconate based solution facilitated neuronal depolarization to examine synaptic inhibition, in [Figure 5](#). It contained (in mM): 125 Cs-gluconate, 10 HEPES, 0.2 EGTA, 2  $\text{MgCl}_2$ , 4 MgATP, 0.4 Tris-GTP and 10  $\text{Na}_2$ -Phosphocreatine, together with 5 mM QX-314 to block  $\text{Na}^+$  channels. Pipette solutions also contained 3 mM biocytin to reveal morphology after recording. They were adjusted to pH 7.3 and osmolality 290 mOsm. Whole-cell, patch-clamp signals were filtered at 3kHz, amplified with a MultiClamp 700B amplifier and acquired with pCLAMP software (Molecular Devices). In a subset of experiments, the following drugs were used to modulate the responses to optogenetic stimulations; the presence of these drugs is indicated in the figure and figure legend, whenever applicable. Monosynaptic excitation induced by optical stimulation was tested in the presence of TTX (1  $\mu\text{M}$ ) and 4-AP (100  $\mu\text{M}$ ). NBQX (10  $\mu\text{M}$ ) and APV (100  $\mu\text{M}$ ) were used to block AMPA and NMDA receptors respectively. Gabazine (10  $\mu\text{M}$ ) was used to block  $\text{GABA}_\text{A}$  receptors. Carbachol (10  $\mu\text{M}$ ) was used to activate acetylcholine receptors. All drugs were bath applied.

## Optical stimulation

LED illumination (Cairn Research, OptoLED) was used to visualize the fluorescent reporters GFP and tdTomato, and to stimulate opsin-expressing axons, using a 470 nm LED for Chronos and a 627 nm LED for Chrimson. Illuminated spots had a diameter of 200  $\mu\text{m}$  with a 60x objective and were centered on the recorded cell soma. Photostimulation thus covered most of the basilar dendrites of layer 3 pyramidal neurons (typical distance from tip of apical to tip of basilar dendrite <400 $\mu\text{m}$ ). Stimuli consisted of light pulses of 0.5 to 5 ms duration, repeated 5-10 times at 20 Hz.

A multiband filter allowed simultaneous stimulation by blue and red LEDs of axons containing Chronos and Chrimson ([Simonnet et al., 2021](#)). Stimulus power intensity (set in mW) was calibrated as light intensity. The response probability of layer 3 cells was calibrated to blue or red illumination of ATN or RSC afferents expressing Chronos or Chrimson to avoid stimulus overlap ([Figure 3—figure supplement 1](#)). Chronos was targeted to the ATN and Chrimson to the RSC

after testing the reverse configuration. Projections from the thalamus are larger and Chronos is more sensitive to blue light, so this configuration assured reliable activation of thalamic fibers at minimal blue-light intensities. For experiments investigating the integration of ATN and RSC inputs, we aimed to give similar weights to both inputs: Chrimson-expressing fibers were stimulated with light of intensity adjusted to initiate optically evoked excitatory postsynaptic potentials (oEPSPs) of amplitude similar to those induced by blue light Chronos-fiber stimuli.

## Data analysis

ATN and RSC projections to the PrS, were analyzed from Chronos and Chrimson expression, using the ImageJ Plot Profile plug-in to quantitate normalized plot profiles (2000 pixels) of horizontal presubicular sections. Dorso-ventral differences were derived by dividing differences in labeling intensity between the PrS and the dentate gyrus (DG) molecular layer by DG intensity in slices from 5 dorso-ventral PrS levels. Values from all animals were averaged and then normalized.

Cells were qualified as synaptically ‘connected’ when they responded to light stimulation of afferent axons with a delay < 8 ms. Slices with very low or absent expression of the fluorescent reporter were excluded. Cells were qualified as ‘non-connected’ synaptically, if they did not respond to light stimulation, but at least one neighboring cell in the same slice did.

Intrinsic neuronal properties were analyzed with custom MATLAB routines to derive 15 electrophysiological parameters (Huang et al., 2017 [↗](#)). Parameters were standardized and unsupervised cluster analysis performed with MATLAB was used to compare different neurons (Huang et al., 2017 [↗](#); Simonnet et al., 2013 [↗](#)).

Axograph was used to analyze responses to optical stimulation of ATN and RSC fibers. Layer 3 neurons were recorded at potentials near -65 mV. Responses to light were averaged from 10 stimulus trains at 20 Hz. Amplitudes and latencies of initial light evoked EPSCs, of latency shorter than 10ms, were quantified from voltage-clamp records. Latency was measured from stimulus onset to 10% of the peak amplitude of the first optically induced EPSC. Amplitude was measured with respect to the pre-stimulus baseline. Paired-pulse ratio (PPR) was defined as the ratio of the amplitude of the second to the first EPSC and 10/1 ratio as that between the 10th and the 1st EPSC, in responses to 20 Hz stimulations. Spike probability was counted over 5 to 10 trials per cell and then averaged over all cells. EPSPs induced by dual wavelength stimulation were analyzed using Axograph and a custom-made software. Events evoked by light stimuli were detected in a window of 1-10 ms after stimulation. EPSP amplitude and integrated area were calculated over 50 ms after stimulation. Baseline suppression was applied using an average of membrane potential during 50 ms before stimulation. Summation of ATN and RSC evoked EPSCs in layer 3 neurons was determined from the amplitude and integral of averaged events. Summation was considered to be supralinear, if values were more than 10% higher than a linear addition, linear for values within  $\pm$  10%, and sublinear for values more than 10% lower.

## Biocytin revelation and morphology

Recorded neurons were filled with biocytin to visualize their morphology and location. Slices were fixed in 4% PFA in 0.1 M PBS at 4°C overnight, washed 3 times in 0.1 PBS and cryoprotected in 30% sucrose. Cell membranes were permeabilized by three freeze-thaw cycles over dry ice and rinsed in PBS. Slices were agitated in a saturation buffer (2% milk, 1% Triton X-100 in 0.1 M PBS) for 2 hours at room temperature. They were then incubated with Streptavidin-Cy5 conjugate (1:500) and DAPI (1:1000) overnight at 4°C. Sections were washed 3 times with PBS and mounted on coverslips with ProLong gold antifade mountant. For anatomical study they were mounted in Mowiol medium. Cell, bead and virus expression were visualized with an epifluorescence microscope. Higher resolution of morphology was obtained with confocal microscopy. Cell morphologies were reconstructed using IMARIS.

## Quantification and statistical analysis

Data was analyzed with AxoGraphX and custom-written software (MATLAB, The MathWorks). Results are given as mean  $\pm$  SEM. Statistical analysis was performed with Prism (GraphPad Software). The Mann-Whitney unpaired t-test was used to compare two groups. The Wilcoxon or Kruskal-Wallis test was used to compare paired groups. Evolution of parameters evoked by optical stimulation (current or potential amplitudes, spike probability) was analyzed with Friedman's test followed by multiple comparison Dunn's test. Šidák's multiple comparison test was also used to compare linear and observed responses to ATN and RSC stimulations. Significance levels are given as p values.

## Data availability

Data are available upon request to the corresponding authors Desdemona Fricker and Louis Richevaux. The Matlab code for analysis of intrinsic properties is available at [https://github.com/schoki0710/Intrinsic\\_Properties](https://github.com/schoki0710/Intrinsic_Properties) .

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## Additional information

### Author contributions

L.R. and D.F. conceived experiments, designed study, and interpreted data. D.F. acquired funding, L.R. and D.F. managed the project. L.R., D.L., M.N., C.M., L.D-R. and N.S-F. performed experiments and analyzed data, I.C. helped with analysis software, L.R. and D.L. prepared figures, L.R. and D.F. wrote the manuscript.

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### **Reviewer #1 (Public review):**

Summary:

In this manuscript, the authors use anatomical tracing and slice physiology to investigate the integration of thalamic (ATN) and retrosplenial cortical (RSC) signals in the dorsal presubiculum (PrS). This work will be of interest to the field, as postsubiculum is thought to be a key region for integrating internal head direction representations with external landmarks. The main result is that ATN and RSC inputs drive the same L3 PrS neurons, which exhibit superlinear summation to near-coincident inputs. Moreover, this activity can induce bursting in L4 PrS neurons, which can pass the signals LMN (perhaps gated by cholinergic input).

#### Strengths:

The slice physiology experiments are carefully done. The analyses are clear and convincing, and the figures and results are well composed. Overall, these results will be a welcome addition to the field.

#### Weaknesses:

The conclusions about the circuit-level function of L3 PrS neurons sometimes outstrip the data, and their model of the integration of these inputs is unclear. I would recommend some revision of the introduction and discussion. I also had some minor comments about the experimental details and analysis.

#### Specific major comments:

(1) I found that the authors' claims sometimes outstrip their data, given that there were no in vivo recordings during behavior. For example, in the abstract that their results indicate "that layer 3 neurons can transmit a visually matched HD signal to medial entorhinal cortex", and in the conclusion they state "[...] cortical RSC projections that carry visual landmark information converge on layer 3 pyramidal cells of the dorsal presubiculum". However, they never measured the nature of the signals coming from ATN and RSC to L3 PrS (or signals sent to downstream regions). Their claim is somewhat reasonable with respect to ATN, where the majority of neurons encode HD, but neurons in RSC encode a vast array of spatial and non-spatial variables other than landmark information (e.g., head direction, egocentric boundaries, allocentric position, spatial context, task history to name a few), so making strong claims about the nature of the incoming signals is unwarranted.

(2) Related to the first point, the authors hint at, but never explain, how coincident firing of ATN and RSC inputs would help anchor HD signals to visual landmarks. Although the lesion data (Yoder et al. 2011 and 2015) support their claims, it would be helpful if the proposed circuit mechanism was stated explicitly (a schematic of their model would be helpful in understanding the logic). For example, how do neurons integrate the "right" sets of landmarks and HD signals to ensure a stable anchoring? Moreover, it would be helpful to discuss alternative models of HD-to-landmark anchoring, including several studies that have proposed that the integration may (also?) occur in RSC (Page & Jeffrey, 2018; Yan, Burgess, Bicanski, 2021; Sit & Goard, 2023). Currently, much of the Discussion simply summarizes the results of the study, this space could be better used in mapping the findings to the existing literature on the overarching question of how HD signals are anchored to landmarks.

#### Comments on revised version:

The authors addressed all of my major points and most of my minor points in the revised submission.

<https://doi.org/10.7554/eLife.92443.2.sa3>

**Reviewer #2 (Public review):**

Richevaux et al investigate how anterior thalamic (AD) and retrosplenial (RSC) inputs are integrated by single presubicular (PrS) layer 3 neurons. They show that these two inputs converge onto single PrS layer 3 principal cells. By performing dual wavelength photostimulation of these two inputs in horizontal slices, the authors show that in most layer 3 cells, these inputs summate supra-linearly. They extend the experiments by focusing on putative layer 4 PrS neurons and show that they do not receive direct anterior thalamic nor retrosplenial inputs; rather, they are (indirectly) driven to burst firing in response to strong activation of the PrS network.

This is a valuable study, which investigates an important question - how visual landmark information (possibly mediated by retrosplenial inputs) converges and integrates with HD information (conveyed by the AD nucleus of the thalamus) within PrS circuitry. The data indicate that near-coincident activation of retrosplenial and thalamic inputs leads to non-linear integration in target layer 3 neurons, thereby offering a potential biological basis for landmark + HD binding.

Main limitations relate to the anatomical annotation of 'putative' PrS L4 neurons, and to the presentation of retrosplenial / thalamic input modularity. Specifically, more evidence should be provided to convincingly demonstrate that the 'putative L4 neurons' of the PrS are not distal subicular neurons (as the authors' anatomy and physiology experiments seem to indicate). The modularity of thalamic and retrosplenial inputs could be better clarified in relation to the known PrS modularity.

<https://doi.org/10.7554/eLife.92443.2.sa2>

**Reviewer #3 (Public review):****Summary:**

The authors sought to determine, at the level of individual presubiculum pyramidal cells, how allocentric spatial information from retrosplenial cortex was integrated with egocentric information from the anterior thalamic nuclei. Employing a dual opsin optogenetic approach with patch clamp electrophysiology, Richevaux and colleagues found that around three quarters of layer 3 pyramidal cells in presubiculum receive monosynaptic input from both brain regions. While some interesting questions remain (e.g. the role of inhibitory interneurons in gating the information flow and through different layers of presubiculum, this paper provides valuable insights into the microcircuitry of this brain region and the role that it may play in spatial navigation.

**Strengths:**

One of the main strengths of this manuscript was that the dual opsin approach allowed the direct comparison of different inputs within an individual neuron, helping to control for what might otherwise have been an important source of variation. The experiments were well-executed and the data rigorously analysed. The conclusions were appropriate to the experimental questions and were well-supported by the results. These data will help to inform in vivo experiments aimed at understanding the contribution of different brain regions in spatial navigation and could be valuable for computational modelling.

**Weaknesses:**

Some attempts were made to gain mechanistic insights into how inhibitory neurotransmission may affect processing in presubiculum (e.g. figure 5) but these



experiments were a little underpowered and the analysis carried out could have been more comprehensively undertaken, as was done for other experiments in the manuscript.

Comments on revised version:

The authors have addressed all of my comments and I have nothing further to add. Well done for an interesting and valuable contribution!

<https://doi.org/10.7554/eLife.92443.2.sa1>

### Author response:

The following is the authors' response to the original reviews.

#### **Reviewer #1 (Public Review):**

##### *Summary:*

*In this manuscript, the authors use anatomical tracing and slice physiology to investigate the integration of thalamic (ATN) and retrosplenial cortical (RSC) signals in the dorsal presubiculum (PrS). This work will be of interest to the field, as the postsubiculum is thought to be a key region for integrating internal head direction representations with external landmarks. The main result is that ATN and RSC inputs drive the same L3 PrS neurons, which exhibit superlinear summation to near-coincident inputs. Moreover, this activity can induce bursting in L4 PrS neurons, which can pass the signals LMN (perhaps gated by cholinergic input).*

##### *Strengths:*

*The slice physiology experiments are carefully done. The analyses are clear and convincing, and the figures and results are well-composed. Overall, these results will be a welcome addition to the field.*

We thank this reviewer for the positive comment on our work.

##### *Weaknesses:*

*The conclusions about the circuit-level function of L3 PrS neurons sometimes outstrip the data, and their model of the integration of these inputs is unclear. I would recommend some revision of the introduction and discussion. I also had some minor comments about the experimental details and analysis.*

##### *Specific major comments:*

*(1) I found that the authors' claims sometimes outstrip their data, given that there were no in vivo recordings during behavior. For example, in the abstract, their results indicate "that layer 3 neurons can transmit a visually matched HD signal to medial entorhinal cortex", and in the conclusion they state "[...] cortical RSC projections that carry visual landmark information converge on layer 3 pyramidal cells of the dorsal presubiculum". However, they never measured the nature of the signals coming from ATN and RSC to L3 PrS (or signals sent to downstream regions). Their claim is somewhat reasonable with respect to ATN, where the majority of neurons encode HD, but neurons in RSC encode a vast array of spatial and non-spatial variables other than landmark information (e.g., head direction, egocentric boundaries, allocentric position, spatial context, task history to name a few), so making strong claims about the nature of the incoming signals is unwarranted.*

We agree of course that RSC does not only encode landmark information. We have clarified this point in the introduction (line 69-70) and formulated more carefully in the abstract (removed the word ‘landmark’ in line 17) and in the introduction (line 82-83). In the discussion we explicitly state that ‘In our slice work we are blind to the exact nature of the signal that is carried by ATN and RSC axons’ (line 522-523).

*(2) Related to the first point, the authors hint at, but never explain, how coincident firing of ATN and RSC inputs would help anchor HD signals to visual landmarks. Although the lesion data (Yoder et al. 2011 and 2015) support their claims, it would be helpful if the proposed circuit mechanism was stated explicitly (a schematic of their model would be helpful in understanding the logic). For example, how do neurons integrate the "right" sets of landmarks and HD signals to ensure stable anchoring? Moreover, it would be helpful to discuss alternative models of HD-to-landmark anchoring, including several studies that have proposed that the integration may (also?) occur in RSC (Page & Jeffrey, 2018; Yan, Burgess, Bicanski, 2021; Sit & Goard, 2023). Currently, much of the Discussion simply summarizes the results of the study, this space could be better used in mapping the findings to the existing literature on the overarching question of how HD signals are anchored to landmarks.*

We agree with the reviewer on the importance of the question, how do neurons integrate the “right” sets of landmarks and HD signals to ensure stable anchoring? Based on our results we provide a schematic to illustrate possible scenarios, and we include it as a supplementary figure (Figure 1, to be included in the ms as Figure 7—figure supplement 2), as well as a new paragraph in the discussion section (line 516-531). We point out that critical information on the convergence and divergence of functionally defined inputs is still lacking, both for principal cells and interneurons

Interestingly, recent evidence from functional ultrasound imaging and electrical single cell recording demonstrated that visual objects may refine head direction coding, specifically in the dorsal presubiculum (Siegenthaler et al. bioRxiv 2024.10.21.619417; doi: <https://doi.org/10.1101/2024.10.21.619417>). The increase in firing rate for HD cells whose preferred firing direction corresponds to a visual landmark could be supported by the supralinear summation of thalamic HD signals and retrosplenial input described in our study. We include this point in the discussion (line 460-462), and hope that our work will spur further investigations.

**Reviewer #2 (Public Review):**

*Richevaux et al investigate how anterior thalamic (AD) and retrosplenial (RSC) inputs are integrated by single presubicular (PrS) layer 3 neurons. They show that these two inputs converge onto single PrS layer 3 principal cells. By performing dual-wavelength photostimulation of these two inputs in horizontal slices, the authors show that in most layer 3 cells, these inputs summate supra-linearly. They extend the experiments by focusing on putative layer 4 PrS neurons, and show that they do not receive direct anterior thalamic nor retrosplenial inputs; rather, they are (indirectly) driven to burst firing in response to strong activation of the PrS network.*

*This is a valuable study, that investigates an important question - how visual landmark information (possibly mediated by retrosplenial inputs) converges and integrates with HD information (conveyed by the AD nucleus of the thalamus) within PrS circuitry. The data indicate that near-coincident activation of retrosplenial and thalamic inputs leads to non-linear integration in target layer 3 neurons, thereby offering a potential biological basis for landmark + HD binding.*

*The main limitations relate to the anatomical annotation of 'putative' PrS L4 neurons, and to the presentation of retrosplenial/thalamic input modularity. Specifically, more evidence should be provided to convincingly demonstrate that the 'putative L4 neurons' of the PrS are not distal subicular neurons (as the authors' anatomy and physiology experiments seem to indicate). The modularity of thalamic and retrosplenial inputs could be better clarified in relation to the known PrS modularity.*

We thank the reviewer for their important feedback. We discuss what defines presubicular layer 4 in horizontal slices, cite relevant literature, and provide new and higher resolution images. See below for detailed responses to the reviewer's comments, in the section 'recommendations to authors'.

### **Reviewer #3 (Public Review):**

#### **Summary:**

*The authors sought to determine, at the level of individual presubiculum pyramidal cells, how allocentric spatial information from the retrosplenial cortex was integrated with egocentric information from the anterior thalamic nuclei. Employing a dual opsin optogenetic approach with patch clamp electrophysiology, Richevaux, and colleagues found that around three-quarters of layer 3 pyramidal cells in the presubiculum receive monosynaptic input from both brain regions. While some interesting questions remain (e.g. the role of inhibitory interneurons in gating the information flow and through different layers of presubiculum, this paper provides valuable insights into the microcircuitry of this brain region and the role that it may play in spatial navigation).*

#### **Strengths:**

*One of the main strengths of this manuscript was that the dual opsin approach allowed the direct comparison of different inputs within an individual neuron, helping to control for what might otherwise have been an important source of variation. The experiments were well-executed and the data was rigorously analysed. The conclusions were appropriate to the experimental questions and were well-supported by the results. These data will help to inform in vivo experiments aimed at understanding the contribution of different brain regions in spatial navigation and could be valuable for computational modelling.*

#### **Weaknesses:**

*Some attempts were made to gain mechanistic insights into how inhibitory neurotransmission may affect processing in the presubiculum (e.g. Figure 5) but these experiments were a little underpowered and the analysis carried out could have been more comprehensively undertaken, as was done for other experiments in the manuscript.*

We agree that the role of interneurons for landmark anchoring through convergence in Presubiculum requires further investigation. In our latest work on the recruitment of VIP interneurons we begin to address this point in slices (Nassar et al., 2024 Neuroscience. doi: 10.1016/j.neuroscience.2024.09.032.); more work in behaving animals will be needed.

### **Reviewer #1 (Recommendations For The Authors):**

*Full comments below. Beyond the (mostly minor) issues noted below, this is a very well-written paper and I look forward to seeing it in print.*

#### **Major comments:**

*(1) I found that the authors' claims sometimes outstrip their data, given that there were no in vivo recordings during behavior. For example, in the abstract, their results indicate "that layer 3 neurons can transmit a visually matched HD signal to medial entorhinal cortex", and in the conclusion they state "[...] cortical RSC projections that carry visual landmark information converge on layer 3 pyramidal cells of the dorsal presubiculum". However, they never measured the nature of the signals coming from ATN and RSC to L3 PrS (or signals sent to downstream regions). Their claim is somewhat reasonable with respect to ATN, where the majority of neurons encode HD, but neurons in RSC encode a vast array of spatial and non-spatial variables other than landmark information (e.g., head direction, egocentric boundaries, allocentric position, spatial context, task history to name a few), so making strong claims about the nature of the incoming signals is unwarranted.*

Our study was motivated by the seminal work from Yoder et al., 2011 and 2015, indicating that visual landmark information is processed in PoS and from there transmitted to the LMN. Based on that, and in the interest of readability, we may have used an oversimplified shorthand for the type of signal carried by RSC axons. There are numerous studies indicating a role for RSC in encoding visual landmark information (Auger et al., 2012; Jacob et al., 2017; Lozano et al., 2017; Fischer et al., 2020; Keshavarzi et al., 2022; Sit and Goard, 2023); we agree of course that this is certainly not the only variable that is represented. Therefore we change the text to make this point clear:

Abstract, line 17: removed the word 'landmark'

Introduction, line 69: added "...and supports an array of cognitive functions including memory, spatial and non-spatial context and navigation (Vann et al., 2009; Vedder et al., 2017)."

Introduction, line 82: changed "...designed to examine the convergence of visual landmark information, that is possibly integrated in the RSC, and vestibular based thalamic head direction signals".

Discussion, line 522-523: added "In our slice work we are blind to the exact nature of the signal that is carried by ATN and RSC axons."

*(2) Related to the first point, the authors hint at, but never explain, how coincident firing of ATN and RSC inputs would help anchor HD signals to visual landmarks. Although the lesion data (Yoder et al., 2011 and 2015) support their claims, it would be helpful if the proposed circuit mechanism was stated explicitly (a schematic of their model would be helpful in understanding the logic). For example, how do neurons integrate the "right" sets of landmarks and HD signals to ensure stable anchoring? Moreover, it would be helpful to discuss alternative models of HD-to-landmark anchoring, including several studies that have proposed that the integration may (also?) occur in RSC (Page & Jeffrey, 2018; Yan, Burgess, Bicanski, 2021; Sit & Goard, 2023). Currently, much of the Discussion simply summarizes the results of the study, this space could be better used in mapping the findings to the existing literature on the overarching question of how HD signals are anchored to landmarks.*

We suggest a physiological mechanism for inputs to be selectively integrated and amplified, based on temporal coincidence. Of course there are still many unknowns, including the divergence of connections from a single thalamic or retrosplenial input neuron. The anatomical connectivity of inputs will be critical, as well as the subcellular arrangement of synaptic contacts. Neuromodulation and changes in the balance of excitation and inhibition will need to be factored in. While it is premature to provide a comprehensive explanation for

landmark anchoring of HD signals in PrS, our results have led us to include a schematic, to illustrate our thinking (Figure 1, see below).

Do HD tuned inputs from thalamus converge on similarly tuned HD neurons only? Is divergence greater for the retrosplenial inputs? If so, thalamic input might pre-select a range of HD neurons, and converging RSC input might narrow down the precise HD neurons that become active (Figure 1). In the future, the use of activity dependent labeling strategies might help to tie together information on the tuning of pre-synaptic neurons, and their convergence or divergence onto functionally defined postsynaptic target cells. This critical information is still lacking, for principal cells, and also for interneurons.

Interneurons may have a key role in HD-to-landmark anchoring. SST interneurons support stability of HD signals (Simonnet et al., 2017) and VIP interneurons flexibly disinhibit the system (Nassar et al., 2024). Could disinhibition be a necessary condition to create a window of opportunity for updating the landmark anchoring of the attractor? Single PV interneurons might receive thalamic and retrosplenial inputs non-specifically. We need to distinguish the conditions for when the excitation-inhibition balance in pyramidal cells may become tipped towards excitation, and the case of coincident, co-tuned thalamic and retrosplenial input may be such a condition. Elucidating the principles of hardwiring of inputs, as for example, selective convergence, will be necessary. Moreover, neuromodulation and oscillations may be critical for temporal coordination and precise temporal matching of HD-to-landmark signals.

We note that matching directional with visual landmark information based on temporal coincidence as described here does not require synaptic plasticity. Algorithms for dynamic control of cognitive maps without synaptic plasticity have been proposed (Whittington et al., 2025, Neuron): information may be stored in neural attractor activity, and the idea that working memory may rely on recurrent updates of neural activity might generalize to the HD system. We include these considerations in the discussion (line 497-501; 521-531) and hope that our work will spur further experimental investigations and modeling work.

While the focus of our work has been on PrS, we agree that RSC also treats HD and landmark signals. Possibly the RSC registers a direction to a landmark rather than comparing it with the current HD (Sit & Goard, 2023). We suggest that this integrated information then reaches PrS. In contrast to RSC, PrS is uniquely positioned to update the signal in the LMN (Yoder et al., 2011), cf. discussion (line 516-520).

*Minor comments:*

*(1) Fig 1 - Supp 1: It appears there is a lot of input to PrS from higher visual regions, could this be a source of landmark signals?*

Yes, higher visual regions projecting to PrS may also be a source of landmark information, even if the visual signal is not integrated with HD at that stage (Sit & Goard 2023). The anatomical projection from the visual cortex was first described by Vogt & Miller (1983), but not studied on a functional level so far.

*(2) Fig 2F, G: Although the ATN and RSC measurements look quite similar, there are no stats included. The authors should use an explicit hypothesis test.*

We now compare the distributions of amplitudes and of latencies, using the Mann-Whitney U test. No significant difference between the two groups were found. Added in the figure legend: 2F, “Mann-Whitney U test revealed no significant difference ( $p = 0.95$ )”. 2G, “Mann-Whitney U test revealed no significant difference ( $p = 0.13$ )”.

*(3) Fig 2 - Supp 2A, C: Again, no statistical tests. This is particularly important for panel A, where the authors state that the latencies are similar but the populations appear to be*

*different.*

Inputs from ATN and RSC have a similar ‘jitter’ (latency standard deviation) and ‘tau decay’. We added in the Fig 2 - Supp 2 figure legend: A, “Mann-Whitney U test revealed no significant difference ( $p = 0.26$ )”. C, “Mann-Whitney U test revealed no significant difference ( $p = 0.87$ )”.

As a complementary measure for the reviewer, we performed the Kolmogorov-Smirnov test which confirmed that the populations’ distributions for ‘jitter’ were not significantly different,  $p = 0.1533$ .

*(4) Fig 4E, F: The statistics reporting is confusing, why are asterisks above the plots and hashmarks to the side?*

Asterisks refer to a comparison between ‘dual’ and ‘sum’ for each of the 5 stimulations in a Sidak multiple comparison test. Hashmarks refer to comparison of the nth stimulation to the 1st one within dual stimulation events (Friedman + Dunn’s multiple comparison test). We mention the two-way ANOVA p-value in the legend (Sum v Dual, for both Amplitude and Surface).

*(5) Fig 5C: I was confused by the 2\*RSC manipulation. How do we know if there is amplification unless we know what the 2\*RSC stim alone looks like?*

We now label the right panel in Fig 5C as “high light intensity” or “HLI”. Increasing the activation of Chrimson increases the amplitude of the summed EPSP that now exceeds the threshold for amplification of synaptic events. Amplification refers to the shape of the plateau-like prolongation of the peak, most pronounced on the second EPSP, now indicated with an arrow. We clarify this also in the text (line 309-310).

*(6) Fig 6D (supplement 1): Typo, "though" should be "through"*

Yes, corrected (line 1015).

*(7) Fig 6G (supplement 1): Typo, I believe this refers to the dotted are in panel F, not panel A.*

Yes, corrected (line 1021).

*(8) Fig 7: The effect of muscarine was qualitatively described in the Results, but there is no quantification and it is not shown in the Figure. The results should either be reported properly or removed from the Results.*

We remove the last sentence in the Results.

*(9) Methods: The age and sex of the mice should be reported. Transgenic mouse line should be reported (along with stock number if applicable).*

We used C57BL6 mice with transgenic background (Ai14 mice, Jax n007914 reporter line) or C57BL6 wild type mice. This is now indicated in the Methods (lines 566-567).

*(10) Methods: If the viruses are only referred to with their plasmid number, then the capsid used for the viruses should be specified. For example, I believe the AAV-CAG-tomato virus used the retroAAV capsid, which is important to the experiment.*



Thank you for pointing this out. Indeed the AAV-CAG-tdTom virus used the retroAAV capsid, (line 575).

(11) *Data/code availability: I didn't see any sort of data/code availability statement, will the data and code be made publicly available?*

Data are stored on local servers at the SPPIN, Université Paris Cité, and are made available upon reasonable request. Code for intrinsic properties analysis is available on github ([https://github.com/schoki0710/Intrinsic\\_Properties](https://github.com/schoki0710/Intrinsic_Properties)). This information is now included (line 717-720).

(12) *Very minor (and these might be a matter of opinion), but I believe "records" should be "recordings", and "viral constructions" should be "viral constructs".*

The text had benefited from proofreading by Richard Miles, who always preferred “records” to “recordings” in his writings. We choose to keep the current wording.

#### **Reviewer #2 (Recommendations For The Authors):**

*Below are two major points that require clarification.*

*(1) In the last set of experiments presented by the authors (Figs 6 onwards) they focus on 'putative L4' PrS cells. For several lines of evidence (outlined below), I am convinced that these neurons are not presubicular, but belong to the subiculum. I think this is a major point that requires substantial clarification, in order to avoid confusion in the field (see also suggestions on how to address this comment at the end of this section).*

*Several lines of evidence support the interpretation that, what the authors call 'L4 PrS neurons', are distal subicular cells:*

*(1.1) The anatomical location of the retrogradely-labelled cells (from mammillary bodies injections), as shown in Figs 6B, C, and Fig. 6, 1B, very clearly indicates that they belong to the distal subiculum. The subicular-to-PrS boundary is a sharp anatomical boundary that follows exactly the curvature highlighted by the authors' red stainings. The authors could also use specific subicular/PrS markers to visualize this border more clearly - e.g. calbindin, Wfs-1, Zinc (though I believe this is not strictly necessary, since from the pattern of AD fibers, one can already draw very clear conclusions, see point 1.3 below).*

Our criteria to delimit the presubiculum are the following: First and foremost, we rely on the defining presence of antero-dorsal thalamic fibers that target specifically the presubiculum and not the neighbouring subiculum (Simonnet et al., 2017, Nassar et al., 2018, Simonnet and Fricker, 2018; Jiayan Liu et al., 2021). This provides the precise outline of the presubicular superficial layers 1 to 3. It may have been confusing to the reviewer that our slicing angle gives horizontal sections. In fact, horizontal sections are favourable to identify the layer structure of the PrS, based on DAPI staining and the variations in cell body size. The work by Ishihara and Fukuda (2016) illustrates in their Figure 12 that the presubicular layer 4 lies below the presubicular layer 3, and forms a continuation with the subiculum (Sub1). Their Figure 4 indicates with a dotted line the “generally accepted border between the (distal) subiculum and PreS”, and it runs from the proximal tip of superficial cells of the PrS toward the white matter, among the radial direction of the cortical tissue. We agree with this definition. Others have sliced coronally (Cembrowski et al., 2018) which renders a different visualization of the border region with the subiculum.

Second, let me explain the procedure for positioning the patch electrode in electrophysiological experiments on horizontal presubicular slices. Louis Richevaux, the first author, who carried out the layer 4 cell recordings, took great care to stay very close (<50 µm)

to the lower limit of the zone where the GFP labeled thalamic axons can be seen. He was extremely meticulous about the visualization under the microscope, using LED illumination, for targeting. The electrophysiological signature of layer 4 neurons with initial bursts (but not repeated bursting, in mice) is another criterion to confirm their identity (Huang et al., 2017). Post-hoc morphological revelation showed their apical dendrites, running toward the pia, sometimes crossing through the layer 3, sometimes going around the proximal tip, avoiding the thalamic axons (Figure 6D). For example the cell in Figure 6, suppl. 1 panel D, has an apical dendrite that runs through layer 3 and layer 1.

Third, retrograde labeling following stereotaxic injection into the LMN is another criterion to define PrS layer 4. This approach is helpful for visualization, and is based on the defining axonal projection of layer 4 neurons (Yoder and Taube, 2011; Huang et al., 2017). Due to the technical challenge to stereotaxically inject only into LMN, the resultant labeling may not be limited to PrS layer 4. We cannot entirely exclude some overflow of retrograde tracers (B) or retrograde virus (C) to the neighboring MMN. This would then lead to co-labeling of the subiculum. In the main Figure 6, panels B and C, we agree that for this reason the red labelled cell bodies likely include also subicular neurons, on the proximal side, in addition to L4 presubicular neurons. We now point out this caveat in the main text (line 324-326) and in the methods (line 591-592).

*(1.2) Consistent with their subicular location, neuronal morphologies of the 'putative L4 cells' are selectively constrained within the subicular boundaries, i.e. they do not cross to the neighboring PrS (maybe a minor exception in Figs. 6\_1D2,3). By definition, a neuron whose morphology is contained within a structure belongs to that structure.*

From a functional point of view, for the HD system, the most important criterion for defining presubicular layer 4 neurons is their axonal projection to the LMN (Yoder and Taube 2011). From an electrophysiological standpoint, it is the capacity of layer 4 neurons to fire initial bursts (Simonnet et al., 2013; Huang et al., 2017). Anatomically, we note that the expectation that the apical dendrite should go straight up into layer 3 might not be a defining criterion in this curved and transitional periarchicortex. Presubicular layer 4 apical dendrites may cross through layer 3 and exit to the side, towards the subiculum (This is the red dendritic staining at the proximal end of the subiculum, at the frontier with the subiculum, Figure 6 C).

*(1.3) As acknowledged by the authors in the discussion (line 408): the PrS is classically defined by the innervation domain of AD fibers. As Figure 6B clearly indicates, the retrogradely-labelled cells ('putative L4') are convincingly outside the input domain of the AD; hence, they do not belong to the PrS.*

The reviewer is mistaken here, the deep layers 4 and 5/6 indeed do not lie in the zone innervated by the thalamic fibers (Simonnet et al., 2017; Nassar et al., 2018; Simonnet and Fricker, 2018) but still belong to the presubiculum. The presubicular deep layers are located below the superficial layers, next to, and in continuation of the subiculum. This is in agreement with work by Yoder and Taube 2011; Ishihara and Fukuda 2016; Boccara, ... Witter, 2015; Peng et al., 2017 (Fig 2D); Yoshiko Honda et al., (Marmoset, Fig 2A) 2022; Balsamo et al., 2022 (Figure 2B).

*(1.4) Along with the above comment: in my view, the optogenetic stimulation experiments are an additional confirmation that the 'putative L4 cells' are subicular neurons, since they do not receive AD inputs at all (hence, they are outside of the PrS); they are instead only indirectly driven upon strong excitation of the PrS. This indirect activation is likely to occur via PrS-to-Subiculum 'back-projections', the existence of which is documented in the literature and also nicely shown by the authors (see Figure 1\_1 and line 109).*

See above. Only superficial layers 1-3 of the presubiculum receive direct AD input.

*(1.5) The electrophysiological properties of the 'putative L4 cells' are consistent with their subicular identity, i.e. they show a sag current and they are intrinsically bursty.*

Presubicular layer 4 cells also show bursting behaviour and a sag current (Simonnet et al., 2013; Huang et al., 2017).

*From the above considerations, and the data provided by the authors, I believe that the most parsimonious explanation is that these retrogradely-labelled neurons (from mammillary body injections), referred to by the authors as 'L4 PrS cells', are indeed pyramidal neurons from the distal subiculum.*

We agree that the retrograde labeling is likely not limited to the presubicular layer 4 cells, and we now indicate this in the text (line 324-326). However, the portion of retrogradely labeled neurons that is directly below the layer 3 should be considered as part of the presubiculum.

*I believe this is a fundamental issue that deserves clarification, in order to avoid confusion/misunderstandings in the field. Given the evidence provided, I believe that it would be inaccurate to call these cells 'L4 PrS neurons'. However, I acknowledge the fact that it might be difficult to convincingly and satisfactorily address this issue within the framework of a revision. For example, it is possible that these 'putative L4 cells' might be retrogradely-labelled from the Medial Mammillary Body (a major subicular target) since it is difficult to selectively restrict the injection to the LMN, unless a suitable driver line is used (if available). The authors should also consider the possibility of removing this subset of data (referring to putative L4), and instead focus on the rest of the story (referring to L3)- which I think by itself, still provides sufficient advance.*

We agree with the reviewer that it is difficult to provide a satisfactory answer. To some extent, the reviewer's comments target the nomenclature of the subicular region. This transitional region between the hippocampus and the entorhinal cortex has been notoriously ill defined, and the criteria are somewhat arbitrary for determining exactly where to draw the line. Based on the thalamic projection, presubicular layers 1-3 can now be precisely outlined, thanks to the use of viral labeling. But the presubicular layer 4 had been considered to be cell-free in early works, and termed 'lamina dissecans' (Boccarda 2010), as the limit between the superficial and deep layers. Then it became of great interest to us and to the field, when the PrS layer 4 cells were first identified as LMN projecting neurons (Yoder and Taube 2011). This unique back-projection to the upstream region of the HD system is functionally very important, closing the loop of the Papez circuit (mammillary bodies - thalamus - hippocampal structures).

We note that the reviewer does not doubt our results, rather questions the naming conventions. We therefore maintain our data. We agree that in the future a genetically defined mouse line would help to better pin down this specific neuronal population.

We thank the reviewer for sharing their concerns and giving us the opportunity to clarify our experimental approach to target the presubicular layer 4. We hope that these explanations will be helpful to the readers of eLife as well.

*(2) The PrS anatomy could be better clarified, especially in relation to its modular organization (see e.g. Preston-Ferrer et al., 2016; Ray et al., 2017; Balsamo et al., 2022). The authors present horizontal slices, where cortical modularity is difficult to visualize and assess (tangential sections are typically used for this purpose, as in classical work*

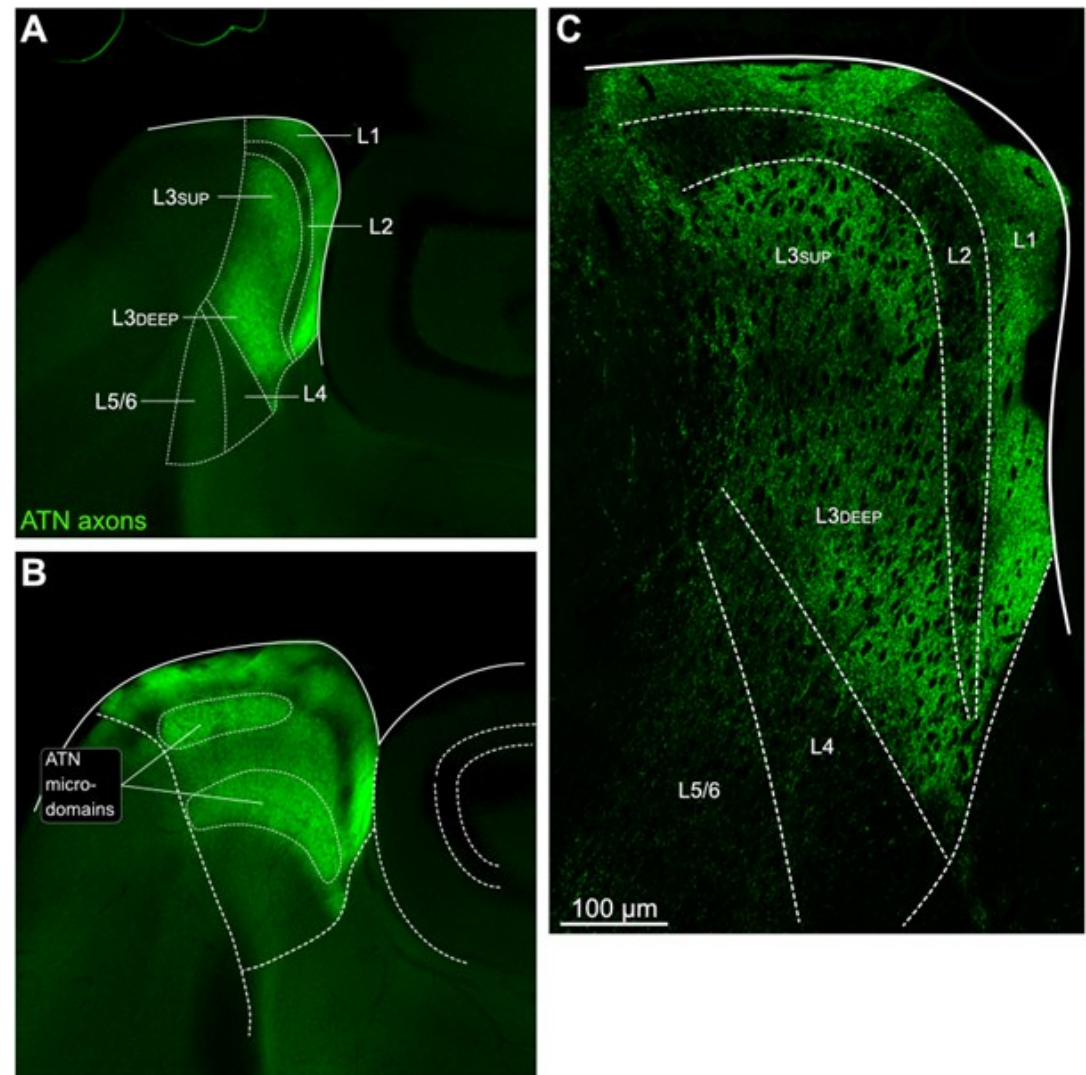
from e.g. barrel cortex). I am not asking the authors to validate their observations in tangential sections, but just to be aware that cortical modules might not be immediately (or clearly) apparent, depending on the section orientation and thickness. The authors state that AD fibers were 'not homogeneously distributed' in L3 (line 135) and refer to 'patches of higher density in deep L3' (line 136). These statements are difficult to support unless more convincing anatomy and . I see some L3 inhomogeneity in the green channel in Fig. 1G (last two panels) and also in Fig. 1K, but this seems to be rather upper L3. I wonder how consistent the pattern is across different injections and at what dorsoventral levels this L3 modularity is observed (I think sagittal sections might be helpful). If validated, these observations could point to the existence of non-homogeneous AD innervation domains in L3 - hinting at possible heterogeneity among the L3 pyramidal cell targets. Notably, modularity in L2 and L1 is not referred to. The authors state that AD inputs 'avoid L2' (line 131) but this statement is not in line with recent work (cited above) and is also not in line with their anatomy data in Fig. 1G, where modularity is already quite apparent in L2 (i.e. there are territories avoided by the AD fibers in L2) and in L1 (see for example the last image in Fig. 1G). This is the case also for the RSC axons (Fig. 1H) where a patchy pattern is quite clear in L1 (see the last image in panel H). Higher-mag pictures might be helpful here. These qualitative observations imply that AD and RSC axons probably bear a precise structural relationship relative to each other, and relative to the calbindin patch/matrix PrS organization that has been previously described. I am not asking the authors to address these aspects experimentally, since the main focus of their study is on L3, where RSC/AD inputs largely converge. Better anatomy pictures would be helpful, or at least a better integration of the authors' (qualitative) observations within the existing literature. Moreover, the authors' calbindin staining in Fig. 1K is not particularly informative. Subicular, PaS, MEC, and PrS borders should be annotated, and higher-resolution images could be provided. The authors should also check the staining: MEC appears to be blank but is known to strongly express *calb1* in L2 (see 'island' by Kitamura et al., Ray et al., Science 2014; Ray et al., frontiers 2017). As additional validation for the staining: I would expect that the empty L2 patches in Figs. 1G (last two panels) would stain positive for Calbindin, as in previous work (Balsamo et al. 2022).

We now provide a new figure showing the pattern of AD innervation in PrS superficial layers 1 to 3, with different dorso-ventral levels and higher magnification (Figure 2). Because our work was aimed at identifying connectivity between long-range inputs and presubicular neurons, we chose to work with horizontal sections that preserve well the majority of the apical dendrites of presubicular pyramidal neurons. We feel it is enriching for the presubicular literature to show the cytoarchitecture from different angles and to show patchiness in horizontal sections. The non-homogeneous AD innervation domains ('microdomains') in L3 were consistently observed across different injections in different animals.

#### Author response image 1.

Thalamic fiber innervation pattern. A, ventral, and B, dorsal horizontal section of the Presubiculum containing ATN axons expressing GFP. Patches of high density of ATN axonal ramifications in L3 are indicated as "ATN microdomains". Layers 1, 2, 3, 4, 5/6 are indicated.

C, High magnification image (63x optical section)(different animal).



We also provide a supplementary figure with images of horizontal sections of calbindin staining in PrS, with a larger crop, for the reviewer to check (Figure 3, see below). We thank the reviewer for pointing out recent studies using tangential sections. Our results agree with the previous observation that AD axons are found in calbindin negative territories (cf Fig 1K). Calbindin+ labeling is visible in the PrS layer 2 as well as in some patches in the MEC (Figure 3 panel A). Calbindin staining tends to not overlap with the territories of ATN axonal ramification. We indicate the inhomogeneities of anterior thalamic innervation that form “microdomains” of high density of green labeled fibers, located in layer 1 and layer 3 (Figure 3, Panel A, middle). Panel B shows another view of a more dorsal horizontal section of the PrS, with higher magnification, with a big Calbindin+ patch near the parasubiculum.

The “ATN+ microdomains” possess a high density of axonal ramifications from ATN, and have been previously documented in the literature. They are consistently present. Our group had shown them in the article by Nassar et al., 2018, at different dorsoventral levels (Fig 1 C (dorsal) and 1D (ventral) PrS). See also Simonnet et al., 2017, Fig 2B, for an illustration of the typical variations in densities of thalamic fibers, and supplementary Figure 1D. Also Jiayan Liu et al., 2021 (Figure 2 and Fig 5) show these characteristic microzones of dense thalamic axonal ramifications, with more or less intense signals across layers 1, 2, and 3. While it is correct that thalamic axons can be seen to cross layer 2 to ramify in layer 1, we maintain that



AD axons typically do not ramify in layer 2. We modify the text to say, “mostly” avoiding L2 (line 130).

The reviewer is correct in pointing out that the 'patches of higher density in deep L3' are not only in the deep L3, as in the first panel in Fig 1G, but in the more dorsal sections they are also found in the upper L3. We change the text accordingly (line 135-136) and we provide the layer annotation in Figure 1G. We further agree with the reviewer that RSC axons also present a patchy innervation pattern. We add this observation in the text (line 144).

It is yet unclear whether anatomical microzones of dense ATN axon ramifications in L3 might fulfill the criteria of a functional modularity, as it is the case for the calbindin patch/matrix PrS organization (Balsamo et al., 2022). As the reviewer points out, this will require more information on the precise structural relationship of AD and RSC axons relative to each other, as well as functional studies. Interestingly, we note a degree of variation in the amplitudes of oEPSC from different L3 neurons (Fig. 2F, discussion line 420; 428), which might be a reflection of the local anatomo-functional micro-organization.

*Minor points:*

*(1) The pattern or retrograde labelling, or at least the way is referred to in the results (lines 104ff), seems to imply some topography of AD-to-PrS projections. Is it the case? How consistent are these patterns across experiments, and individual injections? Was there variability in injection sites along the dorso-ventral and possibly antero-posterior PrS axes, which could account for a possibly topographical AD-to-PrS input pattern? It would be nice to see a DAPI signal in Fig. 1B since the AD stands out quite clearly in DAPI (Nissl) alone.*

Yes, we find a consistent topography for the AD-to-PrS projection, for similar injection sites in the presubiculum. The coordinates for retrograde labeling were as indicated -4.06 (AP), 2.00 (ML) and -2.15 mm (DV) such that we cannot report on possible variations for different injection sites.

*(2) Fig. 2\_2KM: this figure seems to show the only difference the authors found between AD and RS input properties. The authors could consider moving these data into main Fig. 2 (or exchanging them with some of the panels in F-O, which instead show no difference between AD and RSC). Asterisks/stats significance is not visible in M.*

For space reasons we leave the panels of Fig. 2\_2KM in the supplementary section. We increased the size of the asterisk in M.

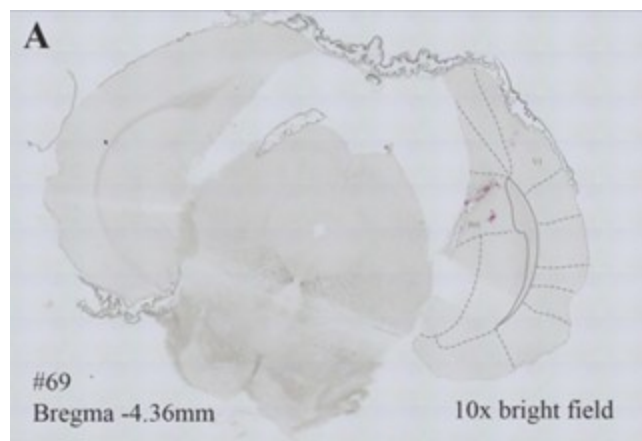
*(3) The data in Fig. 1\_1 are quite interesting, since some of the PrS projection targets are 'non-canonical'. Maybe the authors could consider showing some injection sites, and some fluorescence images, in addition to the schematics. Maybe the authors could acknowledge that some of these projection targets are 'putative' unless independently verified by e.g. retrograde labeling. Unspecific white matter labelling and/or spillover is always a potential concern.*

We now include the image of the injection site for data in Fig. 1\_1 as a supplementary Fig. 1\_2. The Figure 1\_1 shows the retrogradely labeled upstream areas of Presubiculum.



## Author response image 2.

Retrobeads were injected in the right Presubiculum.



*(4) The authors speculate that the near-coincident summation of RS + AD inputs in L3 cells could be a potential mechanism for the binding of visual + HD information in PrS. However, landmarks are learned, and learning typically implies long-term plasticity. As the authors acknowledge in the discussion (lines 493ff) GluR1 is not expressed in PrS cells. What alternative mechanics could the authors envision? How could the landmark-update process occur in PrS, if is not locally stored? RSC could also be involved (Jakob et al) as acknowledged in the introduction - the authors should keep this possibility open also in the discussion.*

A similar point has been raised by Reviewer 1, please check our answer to their point 2. Briefly, our results indicate that HD-to-landmark updating is a multi-step process. RSC may be one of the places where landmarks are learned. The subsequent temporal mapping of HD to landmark signals in PrS might be plasticity-free, as matching directional with visual landmark information based on temporal coincidence does not necessarily require synaptic plasticity. It seems likely that there is no local storage and no change in synaptic weights in PrS. The landmark-anchored HD signals reach LMN via L4 neurons, sculpting network dynamics across the Papez circuit. One possibility is that the trace of a landmark that matches HD may be stored as patterns of neural activity that could guide navigation (cf. El-Gaby et al., 2024, Nature) Clearly more work is needed to understand how the HD attractor is updated on a mechanistic level. Recent work in prefrontal cortex mentions “activity slots” and delineates algorithms for dynamic control of cognitive maps without synaptic plasticity (Whittington et al., 2025, Neuron): information may be stored in neural attractor activity, and the idea that working memory may rely on recurrent updates of neural activity might generalize to the HD system. We include these considerations in the discussion (line 499-503; 523-533) and also point to alternative models (line 518 -522) including modeling work in the retrosplenial cortex.

*(5) The authors state that (lines 210ff) their cluster analysis 'provided no evidence for subpopulations of layer 3 cells (but see Balsamo et al., 2022)' implying an inconsistency; however, Balsamo et al also showed that the (in vivo) ephys properties of the two HD cell 'types' are virtually identical, which is in line with the 'homogeneity' of L3 ephys properties (in slice) in the authors' data. Regarding the possible heterogeneity of L3 cells: the authors report inhomogeneous AD innervation domains in L3 (see also main comment 2) and differences in input summation (some L3 cells integrate linearly, some supra-linearly; lines 272) which by itself might already imply some heterogeneity. I would*

therefore suggest rewording the statements to clarify what the lack of heterogeneity refers to.

We agree. In line 212 we now state “cluster analysis (Figure 2D) provided no evidence for subpopulations of layer 3 cells in terms of intrinsic electrophysiological properties (see also Balsamo et al., 2022).”

(6) *n*=6 co-recorded pairs are mentioned at line 348, but *n*=9 at line 366. Are these numbers referring to the same dataset? Please correct or clarify

Line 349 refers to a set of 6 co-recorded pairs (*n*=12 neurons) in double injected mice with Chronos injected in ATN and Chrimson in RSC (cf. Fig. 7E). The 9 pairs mentioned in line 367 refer to another type of experiment where we stimulated layer 3 neurons by depolarizing them to induce action potential firing while recording neighboring layer 4 neurons to assess connectivity. Line 367 now reads: “In *n* = 9 paired recordings, we did not detect functional synapses between layer 3 and layer 4 neurons.”

### **Reviewer #3 (Recommendations For The Authors):**

*Questions for the authors/points for addressing:*

*I found that the slice electrophysiology experiments were not reported with sufficient detail. For example, in Figure 2, I am assuming that the voltage clamp experiments were carried out using the Cs-based recording solution, while the current clamp experiments were carried out using the K-Gluc intracellular solution. However, this is not explicitly stated and it is possible that all of these experiments were performed using the K-Gluc solution, which would give slightly odd EPSCs due to incomplete space/voltage clamp. Furthermore, the method states that gabazine was used to block GABA(A) receptor-mediated currents, but not when this occurred. Was GABAergic neurotransmission blocked for all measurements of EPSC magnitude/dynamics? If so, why not block GABA(B) receptors? If not blocking GABAergic transmission for measuring EPSCs, why not? This should be stated explicitly either way.*

The addition of drugs or difference of solution is indicated in the figure legend and/or in the figure itself, as well as in the methods. We now state explicitly: “In a subset of experiments, the following drugs were used to modulate the responses to optogenetic stimulations; the presence of these drugs is indicated in the figure and figure legend, whenever applicable.” (line 632). A Cs-based internal solution and gabazine were used in Figure 5, this is now indicated in the Methods section (line 626). All other experiments were performed using K-Gluc as an internal solution and ACSF.

*Methods: The experiments involving animals are incompletely reported. For example, were both sexes used? The methods state "Experiments were performed on wild-type and transgenic C57Bl6 mice" - what transgenic mice were used and why is this not reported in detail (strain, etc)? I would refer the authors to the ARRIVE guidelines for reporting in vivo experiments in a reproducible manner (<https://arriveguidelines.org/>).*

We now added this information in the methods section, subsection “Animals” (line 566-567). Animals of both sexes were used. The only transgenic mouse line used was the Ai14 reporter line (no phenotype), depending on the availability in our animal facility.

*For experiments comparing ATN and RSC inputs onto the same neuron (e.g. Figure 2 supplement 2 G - J), are the authors certain that the observed differences (e.g. rise time and paired-pulse facilitation on the ATN input) are due to differences in the synapses and not a result of different responses of the opsins? Refer to <https://pubmed.ncbi.nlm.nih>*

[.gov/31822522/](https://doi.org/10.7554/eLife.92443.2) from Jess Cardin's lab. This could easily be tested by switching which opsin is injected into which nucleus (a fair amount of extra work) or comparing the Chrimson synaptic responses with those evoked using Chronos on the same projection, as used in Figure 2 (quite easy as authors should already have the data).

We actually did switch the opsins across the two injection sites. In Figure 2 - supplement 2G-J, the values linked by a dashed line result from recordings in the switched configuration with respect to the original configuration (in full lines, Chronos injected in RSC and Chrimson in ATN). The values from switched configuration followed the trend of the main configuration and were not statistically different (Mann-Whitney U test).

*Statistical reporting: While the number of cells is generally reported for experiments, the number of slices and animals is not. While slice ephys often treat cells as individual biological replicates, this is not entirely appropriate as it could be argued that multiple cells from a single animal are not independent samples (some sort of mixed effects model that accounts for animals as a random effect would be better). For the experiments in the manuscript, I don't think this is necessary, but it would certainly reassure the reader to report how many animals/slices each dataset came from. At a bare minimum, one would want any dataset to be taken from at least 3 animals from 2 different litters, regardless of how many cells are in there.*

Our slice electrophysiology experiments include data from 38 successfully injected animals: 14 animals injected in ATN, 20 animals injected in RSC, and 4 double injected animals. Typically, we recorded 1 to 3 cells per slice. We now include this information in the text or in the figure legends (line 159, 160, 297, 767, 826, 831, 832, 839, 845, 901, 941).

*For the optogenetic experiments looking at the summation of EPSPs (e.g. figure 4), I have two questions: why were EPSPs measured and not EPSCs? The latter would be expected to give a better readout of AMPA receptor-mediated synaptic currents. And secondly, why was 20 Hz stimulation used for these experiments? One might expect theta stimulation to be a more physiologically-relevant frequency of stimulation for comparing ATN and RSC inputs to single neurons, given the relevance with spatial navigation and that the paper's conclusions were based around the head direction system. Similarly, gamma stimulation may also have been informative. Did the authors try different frequencies of stimulation?*

Question 1. The current clamp configuration allows to measure EPSP amplification/prolongation by NMDA or persistent Na currents (cf. Fricker and Miles 2000), which might contribute to supralinearity.

Question 2. In a previous study from our group about the AD to PrS connection (Nassar et al., 2018), no significant difference was observed on the dynamics of EPSCs between stimulations at 10 Hz versus 30 Hz. Therefore we chose 20 Hz. This value is in the range of HD cell firing (Taube 1995, 1998 (peak firing rates, 18 to 24 spikes/sec in RSC; 41 spikes/sec in AD)(mean firing rates might be lower), Blair and Sharp 1995). In hindsight, we agree that it would have been useful to include 8Hz or 40Hz stimulations.

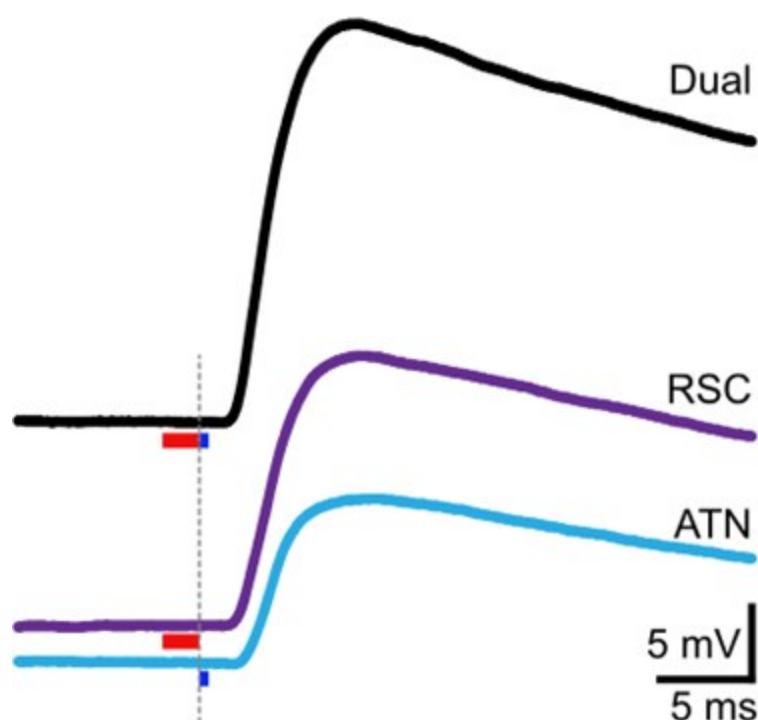
*The GABA(A) antagonist experiments in Figure 5 are interesting but I have concerns about the statistical power of these experiments - n of 3 is absolutely borderline for being able to draw meaningful conclusions, especially if this small sample of cells came from just 1 or 2 animals. The number of animals used should be stated and/or caution should be applied when considering the potential mechanisms of supralinear summation of EPSPs. It looks like the slight delay in RSC input EPSP relative to ATN that was in earlier figures is not present here - could this be the loss of feedforward inhibition?*

The current clamp experiments in the presence of QX314 and a Cs gluconate based internal solution were preceded by initial experiments using puff applications of glutamate to the recorded neurons (not shown). Results from those experiments had pointed towards a role for TTX resistant sodium currents and for NMDA receptor activation as a factor favoring the amplification and prolongation of glutamate induced events. They inspired the design of the dual wavelength stimulation experiments shown in Figure 5, and oriented our discussion of the results. We agree of course that more work is required to dissect the role of disinhibition for EPSP amplification. This is however beyond the present study.

Concerning the EPSP onset delays following RSC input stimulation: In this set of experiments, we compensated for the notoriously longer delay to EPSP onset, following RSC axon stimulation, by shifting the photostimulation (red) of RSC fibers to -2 ms, relative to the onset of photostimulation of ATN fibers (blue). This experimental trick led to an improved alignment of the onset of the postsynaptic response, as shown in the figure below for the reviewer.

### Author response image 3.

In these experiments, the onset of RSC photostimulation was shifted forward in time by -2 ms, in an attempt to better align the EPSP onset to the one evoked by ATN stimulation.



We insert in the results a sentence to indicate that experiments illustrated in Figure 5 were performed in only a small sample of 3 cells that came from 2 mice (line 297), so caution should be applied. In the discussion we formulate more carefully, “From a small sample of cells it appears that EPSP amplification may be facilitated by a reduction in synaptic inhibition (n = 3; Figure 5)” (line 487).

*Figure 7: I appreciate the difficulties in making dual recordings from older animals, but no conclusion about the RSC input can legitimately be made with n=1.*

Agreed. We want to avoid any overinterpretation, and point out in the results section that the RSC stimulation data is from a single cell pair. The sentence now reads : "... layer 4 neurons occurred after firing in the layer 3 neuron, following ATN afferent stimuli, in 4 out of 5 cell pairs. We also observed this sequence when RSC input was activated, in one tested pair." line (347-349)

*Minor points:*

*Line 104: 'within the two subnuclei that form the anterior thalamus' - the ATN actually has three subdivisions (AD, AV, AM) so this should state 'two of the three nuclei that form the anterior thalamus...'*

Corrected, line 103

*Line 125: should read "figure 1F" and not "figure 2F".*

Corrected, line 124

*Line 277-280: Why were two different posthoc tests used on the same data in Figures 3E & F?*

We used Sidak's multicomparison test to compare each event Sum vs. Dual (two different configurations at each time point - asterisks) and Friedman's and Dunn's to compare the nth EPSP amplitude to the first one for Dual events (same configuration between time points - hashmarks). We give two-way ANOVA results in the legend.

<https://doi.org/10.7554/eLife.92443.2.sa0>